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On the Life Cycle Dynamics of Venture-Capital- and Non-Venture-Capital-Financed Firms

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ABSTRACT

We use data over 25 years to understand the life cycle dynamics of VC- and non-VC-financed firms. We find successful and failed VC-financed firms achieve larger scale but are not more profitable at exit than matched non-VC-financed firms. Cumulative failure rates of VC-financed firms are lower, with the difference driven largely by lower failure rates in the initial years after receiving VC. Our results are not driven by VCs disguising failures as acquisitions or by certain types of VCs. The performance difference between VC- and non-VC-financed firms narrows in the post-internet bubble years, but does not disappear.

THE VENTURE CAPITAL (VC) industry has grown dramatically over the past 30 years. In 1980 the total amount of money newly invested by venture capitalists in the U.S. was estimated at $610 million, according to figures published by PricewaterhouseCoopers Moneytree. By 1990 this figure had already increased to $2.3 billion and in 1998 it reached nearly $20 billion. While the numbers fell after the boom period of 1999 and 2000, when VC investment peaked at almost $100 billion, in 2008 the amount invested by venture capitalists was around $28 billion. Clearly, VC has grown as a significant source of financing for new firms over the past 30 years. Yet many questions about VC as an institution, which type of firms it finances and the life cycle dynamics of VC-financed firms

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remain. In particular, much of our understanding about VC-financed firms comes from analysis of firms that are successful and survive. We have little understanding of VC-financed firms that fail, as well as the counterfactual—the life cycle dynamics of firms that could potentially use VC but do not.

Part of the reason that studying these questions is difficult is the scarcity of data on private firms—particularly new non-VC-financed firms. In this paper we use the Longitudinal Business Database (LBD), a panel data set collected by the U.S. Census Bureau that tracks firms from their birth over more than two decades, to address a number of questions related to the role of VC in new firms. Since our data allow large sample identification of firms that do and do not receive VC financing, we are able to characterize and quantify differences between VC-financed and non-VC-financed firms in the early part of their life cycles—from birth to exit—and to shed light on some of the outstanding questions on the role of VC in new firm creation.

We first ask how prevalent VC is in new firm creation. From the Census data we see that a statement on the prevalence of VC depends on the measure used. From the point of view of new firm foundings, we find that VC is close to irrelevant. VC-financed firms are an extremely small percentage of all new firms created—accounting for 0.11% over our 25-year sample period from 1981 to 2005 and increasing to 0.22% over the period 1996–2000. If instead of focusing on the number of firms that get VC backing we focus on employment, we get a different picture of the prevalence of VC in new firm creation and in the economy as a whole. When we measure the amount of employment generated by VC-financed firms, including those that have been exited by their original VC investors, we find that these firms account for between 5.3% to 7.3% of employment in the U.S. during the 2001–2005 period, steadily rising from between 2.7% to 2.8% in the 1981–1985 period. Hence, when measured by employment, VC and the companies it finances are quantitatively sizable.

We examine which kinds of firms receive VC over our sample period. Our results support the notion that VC invests in firms with no immediate revenues. We find that firms started with no commercial revenues are disproportionately financed by VC. This is true in "high-tech" industries, such as biotech and computers, and in "low-tech" industries, such as general retail and wholesale trade. In fact, 47% of new firms in our sample that received VC financing were started without any commercial revenues. In the general population of firms, VC-financed firms are an order of magnitude larger than non-VC-financed firms, as measured by employment and sales. When we examine a sample of VC- and non-VC-financed firms matched on observables at the time the VC-financed firms first receive VC, we also see larger scale after VCs invest. However, we see little difference in measures of profitability between matched VC and non-VC-financed firms before VC-financed firms are exited via acquisition or IPO. These results suggest that the key firm characteristic on which VC focuses is scale or potential for scale, rather than short-term profitability.

Since the Census data allow us to identify which firms fail, or are shut down, we generate statistics on exit percentages for VC-financed firms that cannot be obtained from the standard data sources on VC-financed firms, such as SDC

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Thomson VentureXpert. After the length of a typical VC investment, we find that 39.7% of VC-financed firms fail, 33.5% are acquired, and 16.1% go public. In contrast, 78.9% of non-VC-financed firms fail, 1.04% get acquired, and 0.02% go public.

We next examine a number of hypotheses about firm failure to better understand VC behavior towards companies that do not do well. The characteristics and dynamics of failed firms are arguably the least understood aspect of the VC investing process. Most prior studies focus on VC-financed firms that have exited by IPO or acquisition. For example, in a recent paper Kaplan, Sensoy, and Stromberg (2009) examine the dynamics of VC-financed firms that have gone public from birth to IPO. However, their analysis does not include VC-financed firms that are acquired or that are shut down, nor does it compare VC- to non-VC-financed firms. Our sample allows for in-depth examination of new firm failure dynamics since we observe the eventual outcome of all firms in our sample—failure, acquisition, or IPO—and can examine firm characteristics at the time these outcomes occur.

Using our matched sample of VC-financed and non-VC-financed firms, we first ask whether the larger scale of VC-financed firms reflects a higher failure rate for firms receiving VC. Second, we ask not only whether there is a difference in the overall probability of firm failure but also whether the timing of failure is different between VC- and non-VC-financed firms. Third, we ask if VC-financed firms have different thresholds for failure from non-VC-financed firms. Fourth, we ask if VC-financed firm failures are disguised as acquisitions. Fifth, we ask whether the dynamics of VC- and non-VC-financed firms differ in boom and bust periods. Finally, we ask if the patterns for VC-financed firms reflect the behavior of certain kinds of VC, such as high (low) reputed VC, or whether we see these broad patterns across the board.

In examining our first question, whether the larger scale of VC-financed firms reflects a higher failure rate for firms receiving VC, we find that the cumulative probability of failure is lower for VC-financed firms both in the larger panel and in our matched sample. Thus, it is not the case that the average differences in size between VC- and non-VC-financed firms over time are driven by higher failure rates for VC-financed firms.

We next examine whether the timing of failures differs between VC- and non-VC-financed firms. Historically, many venture capitalists have made very high returns from a few very successful exits, typically through IPOs. Venture capitalists might push their companies hard to grow quickly, deciding relatively rapidly which firms have the best chance of achieving a successful exit and terminating those that do not in the interest of allocating more capital to the likely winners in their portfolios. Or it could be that VCs have a more difficult time judging which firms will be successful in the early stages of investment and equally nurture and invest in all of their firms over a certain period of time. We find that the lower cumulative failure rate of VC-financed firms is driven by a much lower likelihood of failing in the first few years after initially receiving VC. Subsequently, the difference tapers off. These results are consistent with the interpretation that there is a window over which VCs

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allow firms to continue and grow, but once this period is over VCs are just as likely to shut their firms down.

In examining the failure dynamics of VC- and non-VC-financed firms we must also identify which firms get acquired and go public. We use two external data sources to identify acquisitions in our data, the SDC Mergers and Acquisitions (SDC M&A) database and hand searches of news articles obtained from Lexis/Nexis. We find that using Lexis/Nexis news searches to identify acquisitions in the LBD in addition to the SDC M&A database does not have a major impact on the difference in acquisition rates between VC- and non-VC-financed firms, though it does change the overall acquisition rates. Adding the searches increases the set of identified acquisitions for VC-financed firms by 24%, from 2,074 to 2,574 acquisitions. This corresponds to an increase in the total acquisition rate from 20.0% to 24.9%, out of 10,349 VC-financed firms. The Lexis/Nexis searches increase the set of identified acquisitions for non-VC-financed firms by 32%, from 446 to 587 acquisitions. This corresponds to an increase in the total acquisition rate from 4.3% to 5.7%, out of 10,349 non-VC-financed firms.

The third question we ask with regard to failure is if VC-financed firms have different thresholds for failure from non-VC-financed firms. When VC-financed firms are ultimately shut down do they look significantly different in terms of size or profitability relative to non-VC-financed firms that are shut down? Do venture capitalists terminate firms that other investors would keep alive because of better monitoring or higher VC hurdle rates (e.g., Lerner (1995), Barry et al. (1990), and Landier (2003))? Or do venture capitalists give even their failed firms more opportunities to grow and prove themselves relative to investors in non-VC-financed firms in an attempt to learn which of their investments will be huge successes? We find that, in our matched sample, VC-financed firms are significantly larger when they fail in terms of employees and sales and are less profitable at the time of failure. These results suggest that venture capitalists invest heavily in all their firms for an initial period until they have a better sense of which ones will be the successes in their portfolios.

Fourth, we ask if VC-financed firm failures are disguised as acquisitions. While we have observed that VC-financed firms have lower cumulative probabilities of failure, it is possible that venture capitalists are able to sell their poor performers to other companies due to VC connections or natural synergies with potential acquirers, whereas non-VC-financed firms simply must shut down. We examine whether VC-financed firms differ significantly from non-VC-financed firms at the time that they are acquired. We find that there is no significant difference in terms of size, as measured by employment or sales, at the time of acquisition. This is also true when we only examine "going concern" acquisitions, acquisitions for which the acquired firm is not subsequently shut down in the following 2 years. Hence, we do not find evidence that VC failures are being camouflaged as acquisitions. We also examine VC- and non-VC-financed firms at the time they go public and find similar results. VC- and non-VC-financed firms that go public look similar in terms of size when they do go public. This analysis suggests that the large initial investments in employment by VC in the firms it finances is an attempt to get each firm to the

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critical scale and position in which it needs to be in order to successfully exit via IPO or acquisition.

Fifth, we ask if VC-financed firms' outcomes differ in the years following the so-called "internet bubble" of 1998–2000, during which time many firms exited via IPO, when the public equity markets decreased dramatically in value. Is it the case that VC-financed firms overinvested during this period and therefore should perform worse when compared to non-VC-financed firms in the post-bubble period? We find that even when we restrict our analysis to VC-financed firms that first received VC during the bubble years of 1998–2000, VC-financed firms still outperform their matched non-VC-financed counterparts in terms of growth in size, higher IPO and acquisition rates, and lower failure rates.

Prior work by Hsu (2004) suggests that firms are willing to accept worse financing terms in order to be associated with more highly reputed VC firms, and work by Kaplan and Schoar (2005) suggests that some VC firms persistently do better than others in the form of investment returns. Thus, last, but not least, we ask if the patterns for VC-financed firms reflect the behavior of certain kinds of VC, such as high (low) reputed VC, or whether we see these broad patterns across all VCs. We find that the general pattern of failure, acquisition, and IPO that we observe in the overall sample is also observed in the subsamples of both high and low reputed venture capitalists.

Our work adds to the existing literature on the role that VC plays in the firms it finances. To date, empirical studies in this literature mainly focus on VC-financed firms in isolation (e.g., Barry et al. (1990), Gompers and Lerner (2001), Lerner (1995), Kaplan, Sensoy, and Stromberg (2009), Kaplan and Stromberg (2003, 2004)) or on differences between VC- and non-VC-financed firms that have had successful outcomes, such as an IPO or acqusition (e.g., Baker and Gompers (1999), Brav and Gompers (1997), Lindsey (2008), Megginson and Weiss (1991)). A few studies examine smaller hand-collected samples of private non-VC-financed and VC-financed firms but are limited to certain geographies, time periods, industries, and firm outcomes (e.g., Baron, Hannan, and Burton (1999) and Hellmann and Puri (2000, 2002)). In a more recent study Chemmanur, Kartik, and Nandy (2009) use Census Bureau data to find that VC-financed firms are more productive than non-VC-financed firms on average and argue that this arises from the screening and monitoring functions of VCs. Their study uses the Longitudinal Research Database (LRD), which allows them to examine firms' capital and labor inputs in more detail but restricts them to a subsample of older and larger manufacturing firms. Our paper uses the broader-based LBD, which contains the universe of new U.S. employer firms, to understand the dynamics of new VC- and non-VC-financed firms.

The rest of the paper is structured as follows. Section I describes our data. Section II examines which firms receive VC financing. Section III examines differences in the size and exit dynamics of VC- and non-VC-financed firms and explores a number of questions related to firm failure. Section IV concludes.

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I. Data

We first describe the Longitudinal Business Database (LBD), the main panel data set we use to track firms from birth to their first exit event via IPO, acquisition, or shut down. We then describe the other U.S. Census data sets we merge to the LBD to obtain additional information on firm-level sales. We also discuss how we identify VC-financed firms in the data by linking the LBD to VentureSource and VentureXpert, two commonly used commercial data sets on U.S. VC deals. Finally, we describe how we identify IPO and acquisition exits in the data.

A. The Longitudinal Business Database

The LBD is a panel data set that tracks all U.S. employer business establishments from 1975 to the present.[1] The version of the LBD we use ends in 2005. The LBD contains information on employment, payroll, industry, location, and organizational form for each business establishment. We can also observe the years in which an establishment enters and exits the LBD. Since we are interested in understanding the relation between firm-level characteristics and VC financing, we aggregate business establishment-level payroll and employment data for multi-establishment firms using firm-level identifiers. We classify the industry and geography of multi-establishment firms as the modal industry and geography of the firm's business establishments.

For most of our analysis, we begin tracking firms from their birth, or year of first entry into the LBD. A business establishment enters the LBD when it has at least one employee who is paid a wage on which U.S. payroll taxes are levied. Hence, for firms that were started after 1975, we observe them from the point in time at which their first business establishment hires its first tax-paying employee.[2] We classify a firm's year of birth in the LBD as the year in which its first establishment enters the LBD. Most firms, including VC-financed firms, enter the LBD as single-establishment firms.

B. Obtaining Information on Sales

We supplement the payroll and employment data in the LBD with sales data from three additional Census data sources—the Economic Censuses, the Longitudinal Research Database, and the Standard Statistical Establishment List.

[1] The LBD is a business establishment-level data set. A business establishment is part of a firm defined by having a particular geographic location. For example, a law firm with an office in Boston and an office in New York would have two business establishments. Likewise, a manufacturing firm with three different plants operating in different locations, for example, two in Illinois and one in Wisconsin, would have three business establishments. For a detailed description of the variables contained in the LBD and how it is formed, see Jarmin and Miranda (2002).

[2] Firms whose only employees are their owners are still included in the LBD as long as the owners pay themselves some level of wages.

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The Economic Censuses of Services, Retail Trade, and Wholesale Trade collect information on the value of goods produced in each business establishment in the services, retail trade, and wholesale trade industries every 5 years, for years ending in 2 and 7. For firms in the manufacturing industries, we obtain information on the value of goods produced in the Economic Census of Manufactures, also collected in years ending 2 and 7, as well as the Annual Survey of Manufactures. The Economic Census of Manufactures combined with the Annual Survey of Manufactures in the non-Census years comprise the Longitudinal Research Database (LRD). Because the Annual Survey of Manufactures oversamples larger firms, using this database for sales data slightly tilts our firms with sales data towards larger and more successful firms. However, this is true for both VC- and non-VC-financed firms, and only relevant for manufacturing firms.

We use the Standard Statistical Establishment List (SSEL) to obtain revenue data reported on the tax returns for a subset of firms in the LBD. The SSEL is a list of business establishments maintained by the U.S. Census Bureau that it uses as the basis for the Economic Census and related surveys. The SSEL contains data from U.S. government administrative records, such as tax returns. Beginning in 1995, the SSEL contains firm-level revenues as reported on firms' tax returns. Tax return revenue data are available for about two-thirds of firms in the SSEL. The advantage of the SSEL tax return data is that it gives a measure of actual cash obtained by each firm in a given year, as opposed to an estimate of the value of goods produced as in the Economic Census. The disadvantage, however, is that the data are only available in the last 11 years of our 25-year sample period. Thus, we primarily use the sales data in the Economic Census and the LRD in our empirical analysis.

C. Identifying VC-Financed Firms in the LBD

We identify firms in the LBD as VC-financed if they can be matched to extracts of VentureSource and VentureXpert, which identify firms that receive their first rounds of VC financing between 1981 and 2005. VentureSource and VentureXpert are proprietary databases maintained and sold by Dow Jones and Thomson Financial and contain information on venture capital investment firms and the companies in which they invest. Both VentureSource and VentureXpert contain information on when rounds of VC financing occur, which VC funds participate in the rounds, and how much financing was received in the rounds. VentureXpert includes both VC and buyout deals, so we only include in our extract any VC-financed firm located in the United States whose first round of financing is classified as "Startup/Seed," "Early Stage," "Expansion," or "Later Stage." We also exclude companies that are missing name or address information in both VentureSource and VentureXpert. We match firms in the union of VentureXpert and VentureSource using name and address matching. Section A of Appendix A contains a detailed description of how we match VentureSource and VentureXpert to the LBD.

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D. Matching VC-Financed and Non-VC-Financed Firms in the LBD

We also form a one-to-one matched sample of VC- and non-VC-financed firms in the LBD based on firm characteristics at the time VC-financed firms first receive VC. One may ask whether differences between VC- and non-VC-financed firms are due to venture capitalists selecting better firms or whether the nature of VC financing itself and the role venture capitalists play in new firms directly cause the observed differences between VC- and non-VC-financed firms. Hence, our next step is to match each VC-financed firm to a non-VC-financed firm at the time of getting VC funding based on four characteristics, namely, age of the firm, four-digit SIC code, geographical region, and employment size.[3] We re-examine the relation between VC financing and firm size and exit for a set of firms that are observationally similar at the time one of them gets VC funding and the other does not. While this does not allow us to completely distinguish selection from causation, it does allow us to make a statement about differences between VC- and non-VC-financed firms that are identical on certain observable characteristics at the time one of the firms receives VC financing.

Section B of Appendix A contains a description of the matching process as well as summary statistics for the matched sample. The final matched sample contains 10,349 VC-financed and 10,349 non-VC-financed firms that enter the LBD between 1981 and 2005. In Section C of Appendix A, we compare the characteristics of the 10,349 VC-financed firms in the one-to-one matched sample to all other VC-financed firms in VentureSource and VentureXpert. We find that the VC-financed firms in the matched sample are slightly smaller and less likely to have an IPO and are more likely to fail when compared to other VC-financed firms. Thus, it does not appear to be the case that our findings with regard to differences in the size and exit dynamics of VC- and non-VC-financed firms in the matched sample are biased by examining only successful VC-financed firms.

E. Identifying Firm Exits

For most of our analysis, we track firms from their year of entry to the year of their first exit event—IPO, acquisition, or shut down—or until 2005 when our data end. We classify a firm as shutting down, or failing, when all of its establishments exit the LBD. The year of failure for the firm will be the year in which the last establishment is shut down. We identify acquisitions in the LBD in several ways. First, we classify a firm as having been acquired if all of its business establishments have been acquired by another firm as identified in the LBD. However, ownership changes in the LBD are tracked primarily in the years in which the Economic Census is taken. Since the Economic Census only occurs every 5 years, some ownership changes may not be recorded if acquired establishments are shut down before the next Economic Census is taken. We therefore also use two external data sources to identify acquisitions in the LBD.

[3] We do not match on sales because we only observe this variable in 5-year intervals over our sample period for most industries.

We first merge the SDC Platinum Mergers and Acquisitions (SDC M&A) database to the LBD to identify acquisition targets. If a firm in the LBD is matched to an acquisition target in SDC, we also classify the firm as having been acquired. Because the SDC M&A database has greater coverage of larger acquisitions and acquisitions by public acquirers, we also perform hand searches of news articles published between 1981 and 2005 as obtained from Lexis/Nexis to identify additional acquisition events. Appendix B describes our identification of acquisitions in the LBD in greater detail. Using Lexis/Nexis news searches to identify acquisitions in the LBD in addition to the SDC M&A database does not have a major impact on the difference in acquisition rates between VC- and non-VC-financed firms, though it does change the overall acquisition rates. Adding the searches increases the set of identified acquisitions for VC-financed firms by 24%, from 2,074 to 2,574 acquisitions. This corresponds to an increase in the total acquisition rate from 20.0% to 24.9%, out of 10,349 VC-financed firms. The Lexis/Nexis searches increase the set of identified acquisitions for non-VC-financed firms by 32%, from 446 to 587 acquisitions. This corresponds to an increase in the total acquisition rate from 4.3% to 5.7%, out of 10,349 non-VC-financed firms.

We identify firms that have had an IPO by merging the LBD to a list of firms having IPOs between 1981 and 2005 in the United States taken from the SDC Platinum Global New Issues database.

II. Which Firms Receive VC Financing?

How prevalent is VC in our data? Table I shows that a statement on the prevalence of VC depends on the measure used. In Table I, Panel A, we see that, from the point of view of the number of firms having ever received VC, VC is close to irrelevant. Over the entire 25-year sample period from 1981 to 2005, on average only 0.10% of firms per year have ever received VC financing. This percentage includes firms that received VC between 1981 and 2005 and subsequently went public or were acquired by a larger firm, provided at least one establishment of the acquired VC-financed firm remains an active part of the combined firm. The percentage of firms having ever received VC increases over the sample period from 0.05% in the 1981–1985 period to 0.16% in the 2001–2005 period, reflecting the growth in the VC markets. However, even in more recent years VC-financed firms account for only a very small fraction of all firms in existence.

If instead of focusing on the number of firms that get VC financing we focus on employment by VC-financed firms, we obtain a different picture of the prevalence of VC in the economy. Panel B of Table I considers the amount of employment generated by the VC-backed firms in Panel A. We find that employment generated by firms that have ever received VC, including otherwise non-VC-financed firms that have active acquired VC-financed establishments, accounts for 5.5% of employment per year in the United States over our entire sample. This percentage rises steadily from 2.8% in the 1981–1985 period to 7.3% in the 2001–2005 period. We also report firm counts and employment

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Table I

The Role of VC in Total Employment and New Firm Creation, 1981–2005

The underlying data are taken from the Longitudinal Business Database (LBD) for the years 1981–2005. Panel A reports the yearly number of firms in existence and the percentage of these firms that have ever received VC. Panel B reports the yearly total employment and the percentage represented by firms that have ever received VC. In Panels A and B, VC-financed firms are those that have ever received VC at any point between 1981 and 2005 and remain in the sample after the VC investors have exited. In the first row of Panels A and B, VC-financed firms that are exited by IPO or by acquisition both remain in the sample after exit, as long as in the case of acquisition at least one VC-financed target establishment remains part of the acquiring firm. In the second row of Panels A and B, only VC-financed firms that are exited by IPO remain in the sample after exit. Also included in Panels A and B are firms that have not yet received VC but will do so at some point between 1981 and 2005. Panel C reports the total number of new firms created between 1981 and 2005, that is, firms that enter the LBD for the first time between 1981 and 2005, and the percentage of these firms that receive VC, either when they are created or in a subsequent year up to 2005.

	Time Period					
	1981–2005	1981–1985	1986–1990	1991–1995	1996–2000	2001–2005
Panel A. Firms That Have Ever Received VC						
% of firms in existence that have ever received VC (including firms with acquired VC-financed establishments)	0.10%	0.05%	0.08%	0.09%	0.14%	0.16%
% of firms in existence that have ever received VC (excluding firms with acquired VC-financed establishments)	0.09%	0.05%	0.08%	0.08%	0.13%	0.14%
Average number of firms in existence	5,955,132	4,980,533	5,579,077	6,084,340	6,453,045	6,678,664
Panel B. Employment by Firms That Have Ever Received VC						
% of total employment by firms in existence that have ever received VC (including firms with acquired VC-financed establishments)	5.5%	2.8%	4.6%	5.7%	6.8%	7.3%
% of total employment by firms in existence that have ever received VC (excluding firms with acquired VC-financed establishments)	4.0%	2.7%	3.7%	3.8%	4.2%	5.3%
Average total employment by firms in existence	112,034,832	90,903,883	103,437,645	110,974,117	124,145,935	130,712,580
Panel C. VC and New Firm Creation						
% of new firms that receive VC	0.11%	0.08%	0.06%	0.08%	0.22%	0.10%
Total number of new firms	16,304,287	2,900,686	3,415,869	3,191,949	3,209,296	3,586,487

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statistics for the sample of VC-financed firms when we exclude otherwise non-VC-financed firms that have acquired active VC-financed establishments from the sample. This number provides a lower bound on the prevalence of VC financing in the economy. Even considering this more conservative definition of VC-financed firms, we see that VC-financed firms account for 4.0% of employment over the entire 1981–2005 period, rising from 2.7% in the 1981–1985 period to 5.3% in the 2001–2005 period.

We also find that VC-financed firms are an extremely small percentage of all new firms created in the LBD. In Panel C of Table I we see that 0.11% of all new firms created during the 1981–2005 period ever receive VC financing. This percentage peaks at 0.22% of all new firms created in the 1996–2000 period. The results in Table I suggest that VC finances firms that grow rapidly and eventually become large players in the economy. We next examine what is different about the new firms that receive VC financing compared to those that do not and what criteria venture capitalists might consider when making their investments in firms.

A. VC Financing by Industry

While we have some sense that VC-financed firms are concentrated in certain high-tech industries from surveys such as Moneytree and aggregate statistics reported by the National Venture Capital Association, it is unclear how the industry composition of firms that receive VC compares to the industry composition of firms that do not. We start by asking what industries VC backs relative to other sources of entrepreneurial capital over our 25-year sample period using Census data. Table II presents the average yearly number of firms in existence, their total employment by industry, and the percentages that are represented by firms that have ever received VC. The underlying sample is the same as that in Panels A and B of Table I. We classify firms into nine industry categories that correspond to the industry categories used by VentureSource and VentureXpert. In certain industries, such as Computers, Electronics, and Telecom, the percentage of firms that have ever received VC and the total employment represented by these firms is much higher than the economy-wide average. In these industries, if we include firms that have acquired VC-financed establishments that remain active parts of the acquiring firm, the percentage of firms having ever received VC exceeds 1% over the entire sample period, at 1.04% for Computers, 1.20% for Electronics, and 1.64% for Telecom. The percentage of firms having received VC is 0.95% for Computers, 1.04% for Electronics, and 1.43% for Telecom if we exclude firms that have aquired VC-financed establishments. The percentage of total employment by VC-financed firms in Computers, Electronics, and Telecom is around 20% if we include firms that have acquired VC-financed establishments in our set of VC-financed firms, at 19.0% for Computers, 20.4% for Electronics, and 21.1% for Telecom. If we exclude firms that have acquired VC-financed establishments from our set of VC-financed firms, we find that VC-financed firms account for 14.5% of employment in the Computers industry, 14.7% of

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Table II

The Role of VC in Total Employment by Industry, 1981–2005

The underlying data are taken from the Longitudinal Business Database (LBD) for the years 1981–2005. The table reports the yearly number of firms in existence and the yearly total employment by industry. Also reported are the percentages of these two figures that are represented by firms that have ever received VC. In Panels A and B, VC-financed firms are those that have ever received VC at any point between 1981 and 2005 and remain in the sample after the VC investors have exited. In the first row of Panels A and B, VC-financed firms that are exited by IPO or by acquisition both remain in the sample after exit, as long as in the case of acquisition at least one VC-financed target establishment remains part of the acquiring firm. In the second row of Panels A and B, only VC-financed firms that are exited by IPO remain in the sample after exit. Also included in Panels A and B are firms that have not yet received VC but will do so at some point between 1981 and 2005. Industries are selected to correspond to the industrial classifications given to VC-financed firms by VentureXpert and VentureSource. SIC codes are assigned to these nine industries based on the observed mapping of the industrial classifications to VC-financed firms' primary SIC codes in the LBD. A firm is in the "Computer" industry if its primary SIC code is 3570–5379, 5044, 5045, 5734, or 7370–7379. A firm is in the "Biotech/Medical" industry if its primary SIC code is 2830–2839, 3826, 3841–3851, 5047, 5048, 5122, 6324, 7352, 8000–8099, or 8730–8739 excluding 8732. A firm is in the "Electronics" industry if its primary SIC code is 3600–3629, 3643, 3644, 3670–3699, 3825, 5065, or 5063. A firm is in the "Telecom" industry if its primary SIC code is 3660–3669 or 4810–4899. A firm is in the "Consumer Goods" industry if its primary SIC code is 2310–2325, 2329, 2331–2342, 2360–2389, 2392, 2510–2519, 2844, 3140–3149, 3630–3639, 3931, 3942, 3944, 3946, 5023, 5064, 5091, 5092, 5094, 5136, 5137, 5139, 5140–5149, 5180, 5181, 5182, 5192, 5194, 5199, 5411, 5421, 5431, 5441, 5451, 5499, 5531, 5610–5699, 5710–5731, 5735, 5736, 5812, 5183, 5910–5963, 5992, 5993, 5994, or 5999. A firm is in the "Finance" industry if its primary SIC code is 6020–6062, 6090–6099, 6111–6289, 6311, 6321, 6331, 6351, 6361, 6411, 6510–6553, 6712,6722, 6726, or 6790–6799. A firm is in the "Business Services" industry if its primary SIC code is 7310–7349 or 8710–8748. A firm is in the "Industrial Goods" industry if its primary SIC code is 1311, 1381, 1382, or 1389, or if its primary SIC code is a manufacturing SIC code, 2010–3999, not already used to define the previous seven industries. A firm is in the "Other" industry if its primary SIC code is not used to define any of the previous eight industries.

	All Industries	Computer	Biotech/ Medical	Electronics	Telecom	Finance	Consumer Goods	Business Services	Industrial Goods	Other
Panel A. Firms That Have Ever Received VC 1981–2005										
% of firms in existence that have ever received VC (including firms with acquired VC-financed establishments)	0.10%	1.04%	0.17%	1.20%	1.64%	0.04%	0.04%	0.11%	0.20%	0.03%

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% of firms in existence that have ever received VC (excluding firms with acquired VC-financed establishments)	0.09%	0.95%	0.16%	1.04%	1.43%	0.03%	0.03%	0.10%	0.17%	0.03%
Average number of firms in existence	5,955,132	203,914	475,227	37,231	17,577	483,036	1,063,667	313,135	310,537	3,073,609
Panel B. Employment by Firms That Have Ever Received VC 1981–2005										
% of total employment by firms in existence that have ever received VC (including firms with acquired VC-financed establishments)	5.5%	19.0%	5.4%	20.4%	21.1%	6.8%	6.1%	4.9%	3.2%	3.5%
% of total employment by firms in existence that have ever received VC (excluding firms with acquired VC-financed establishments)	4.0%	14.5%	4.0%	14.7%	14.4%	5.1%	4.5%	3.4%	2.2%	2.5%
Average total employment by firms in existence	112,034,832	4,353,253	11,214,755	1,898,511	1,635,223	7,807,423	19,836,900	3,630,903	10,292,455	51,365,410

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employment in the Electronics industry, and 14.4% of employment in the Telecom industry.

Table III presents industry counts for all new VC- and non-VC-financed firms in the LBD. Panel A of Table III reports industry counts for all firms that first enter the LBD between 1981 and 2005. Consistent with Table II, we observe that in some industries the proportion of VC-financed firms being created is much higher. In particular, the percentage of VC-financed firms created in the Computer, Electronics, and Telecom industries together is 0.8%, almost eight times greater than the population average. The percentage of VC-financed firms created in the Biotech/Medical industry is 0.3%.[4] This breakdown is consistent with the notion that VC disproportionately backs firms in high-tech industries.

However, Table III also demonstrates that VC finances a large number of new firms in low-tech industries as well, although as a much smaller percentage of the total number of new firms in these industries. Over the 1981–2005 period, VC financed 17,763 new firms, of which 11,286 (or 64%) are in the high-tech industries of Computer, Biotech/Medical, Electronics, and Telecom, and 6,477 (or 36%) are in the low-tech industries of Finance, Consumer Goods, Business Services, Industrial Goods, and Other.

Panel B of Table III reports industry counts for VC- and non-VC-financed firms created between 1995 and 2005, the period over which we have tax return revenue data from the SSEL. Panel C reports the percentage of new VC- and non-VC-financed firms created in each industry between 1995 and 2005 that have zero revenues in their first year. We find that 47% of VC-financed firms created between 1995 and 2005 had zero revenues in their first year versus 9% of non-VC-financed firms. Thus, VC disproportionately finances firms that are created without having any initial commercial revenues. Even in the low-tech industries, the percentage of new VC-financed firms that have zero commercial revenues in their first year remains high, at between 36% and 42% of firms. The percentage of new VC-financed firms in the high-tech industries of Biotech/Medical, Electronics, and Telecom that are started with no revenues is even higher at between 58% and 68%.

Table III suggests that a large percentage of the firms that venture capitalists back develop new products without any initial sales. Insofar as these are more likely to be idea based or innovator firms, this finding is consistent with prior work by Kortum and Lerner (2000), who assess the contribution of VC to innovation in the U.S. and find that it is positively related, and Hellmann and Puri (2000), who ask if firms with an innovator strategy are more likely to get VC.

[4] These percentages are obtained by dividing the number of new VC-financed firms by the total number of firms created in an industry, reported in Table III, Panel A. For the Computer, Telecom, and Electronics industries the percentage is calculated as (6,995 + 971 + 948)/(989,947 + 73,612 + 54,577). For the Biotech/Medical industry, the percentage is calculated as 2,372/878,773.

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Table III

The Role of VC in New Firm Creation by Industry, 1981–2005

The sample is comprised of all firms that enter the Longitudinal Business Database (LBD) for the first time between 1981 and 2005 and is the same as the sample described in Panel C of Table I. VC-financed firms are firms that receive VC when they are created or in a subsequent year up to 2005. Panel A presents the total number of new firms by industry for VC- and non-VC-financed firms. The numbers in parentheses are the percentages of the total number of new VC- or non-VC-financed firms in an industry. Panel B presents the total number of new firms by industry for VC- and non-VC-financed firms that enter the LBD for the first time between 1995 and 2005. The numbers in parentheses are the percentages of the total number of VC- or non-VC-financed firms in an industry. Panel C reports the percentage of VC and non-VC-financed firms in each industry that have zero revenues in their first year. Revenue data are taken from the Standard Statistical Establishment List (SSEL). SSEL revenue data are missing for one third of non-VC-financed firms and for one quarter of VC-financed firms. Industries are selected to correspond to the industrial classifications given to VC-financed firms by VentureXpert and VentureSource. SIC codes are assigned to these nine industries based on the observed mapping of VC-financed firms' industry classifications to their primary SIC codes in the LBD. This mapping is described in the caption of Table II.

	All Industries	Computer	Biotech /Medical	Electronics	Telecom	Finance	Consumer Goods	Business Services	Industrial Goods	Other
Panel A. New Firms Created between 1981 and 2005										
Total number of new VC-financed firms	17,763	6,995	2,372	971	948	617	1,049	1,408	1,178	2,225
		(39%)	(13%)	(5%)	(5%)	(3%)	(6%)	(8%)	(7%)	(13%)
Total number of new non-VC-financed firms	16,286,524	982,952	876,401	72,641	53,629	1,175,770	2,889,916	1,058,628	628,148	8,548,439
		(6%)	(5%)	(0.4%)	(0.3%)	(7%)	(18%)	(7%)	(4%)	(53%)
Panel B. New Firms Created between 1995 and 2005										
Total number of new VC-financed firms	11,482	5,251	1,262	513	619	414	717	996	519	1,191
		(46%)	(11%)	(5%)	(5%)	(4%)	(6%)	(9%)	(5%)	(10%)
Total number of new non-VC-financed firms	7,442,899	719,953	427,841	29,095	28,060	615,418	1,313,973	592,753	240,904	3,474,902
		(10%)	(6%)	(0.4%)	(0.4%)	(8%)	(18%)	(8%)	(3%)	(47%)
Panel C. Percentage of New Firms Created between 1995 and 2005 with Zero Revenues in First Year										
New VC-financed firms	47%	45%	68%	58%	59%	37%	42%	36%	40%	38%
New non-VC-financed firms	9%	8%	10%	8%	10%	25%	6%	6%	7%	9%

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B. VC Financing and Firm Size

We next examine whether VC appears to have scale (actual or potential) criteria for the firms in which it invests.

B.1. Comparing All VC- and Non-VC-Financed Firms

Figures 1a and 1b depict the average employment and sales by firm age for all VC- and non-VC-financed firms in the LBD created between 1981 and 2005. Figure 1c plots our measure of profitability, (sales-payroll)/sales, or payroll margin. We do not observe other firm cost variables, so we only use employment costs in this measure. The sales variable depicted in Figures 1b and 1c is the sales variable from the Economic Censuses of Services, Wholesale Trade, and Retail Trade and the LRD. In the remaining analysis we use this measure of firm sales, rather than SSEL tax return revenues, because we can observe this measure of sales over our entire sample period. In Figures 1a, 1b, and 1c, we track firms until their first exit event, that is, failure, acquisition, or IPO, if applicable, since we are only interested in examining firms for which VCs are currently invested and involved.

Figures 1a and 1b show that VC-financed firms are larger than non-VC-financed firms, measured by both employment and sales, at each age of the life cycle prior to first exit. In addition, the size difference between VC- and non-VC-financed firms becomes larger with firm age, that is, the average growth rate of VC-financed firms is larger. This suggests that actual or potential scale of employment and sales is an important criterion in how venture capitalists choose which firms to finance. Figure 1c shows that another difference between VC- and non-VC-financed firms is that VC-financed firms spend a much larger share of their sales on payroll expenses for employees.

B.2. Comparing Matched VC- and Non-VC-Financed Firms

Figures 2a and 2b plot average firm employment and sales in "match time," or years from matching, for the sample of one-to-one matched VC-financed and non-VC-financed firms. We see that, at the point of matching, VC-financed and non-VC-financed firms have similar employment, by construction of the matching algorithm, and sales levels.

After VC financing, we see very rapid growth in the employment of VC-financed firms relative to non-VC-financed firms. While VC-financed and non-VC-financed firms are matched at an average of 26 employees each, 3 years later VC-financed firms have on average 55 employees while non-VC-financed firms have 38 employees. We continue to see greater employment growth by VC-financed firms relative to non-VC-financed firms as time goes by. In the first several years after matching in Figure 2b, VC-financed firms and non-VC-financed firms both experience increases in sales, but sales growth is greater for VC-financed firms. Non-VC-financed firms do not "catch up" to VC-financed firms in these later years in terms of employment or sales. Non-VC-financed

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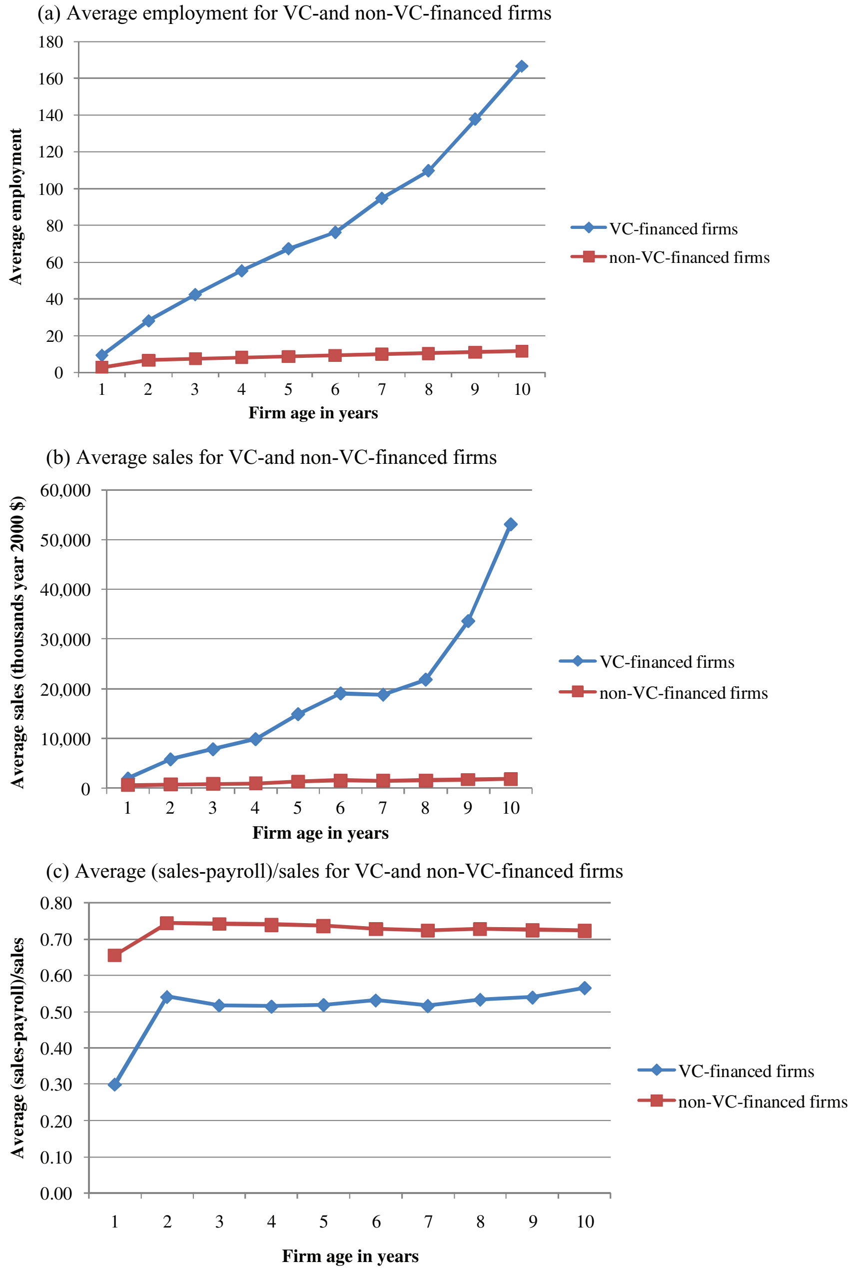

Figure 1. Average size and profitability of all VC- and non-VC-financed firms created between 1981 and 2005. The sample consists of all firms that first enter the Longitudinal Business Database (LBD) between 1981 and 2005. The firms are tracked from the time they enter the LBD to the year of their first exit event, if applicable, via failure, acquisition, or IPO. Employment and payroll data are taken from the LBD. Sales data are taken from the 1982, 1987, 1992, 1997, and 2002 waves of the Censuses of Services, Retail Trade, and Wholesale Trade and from the Longitudinal Research Database.

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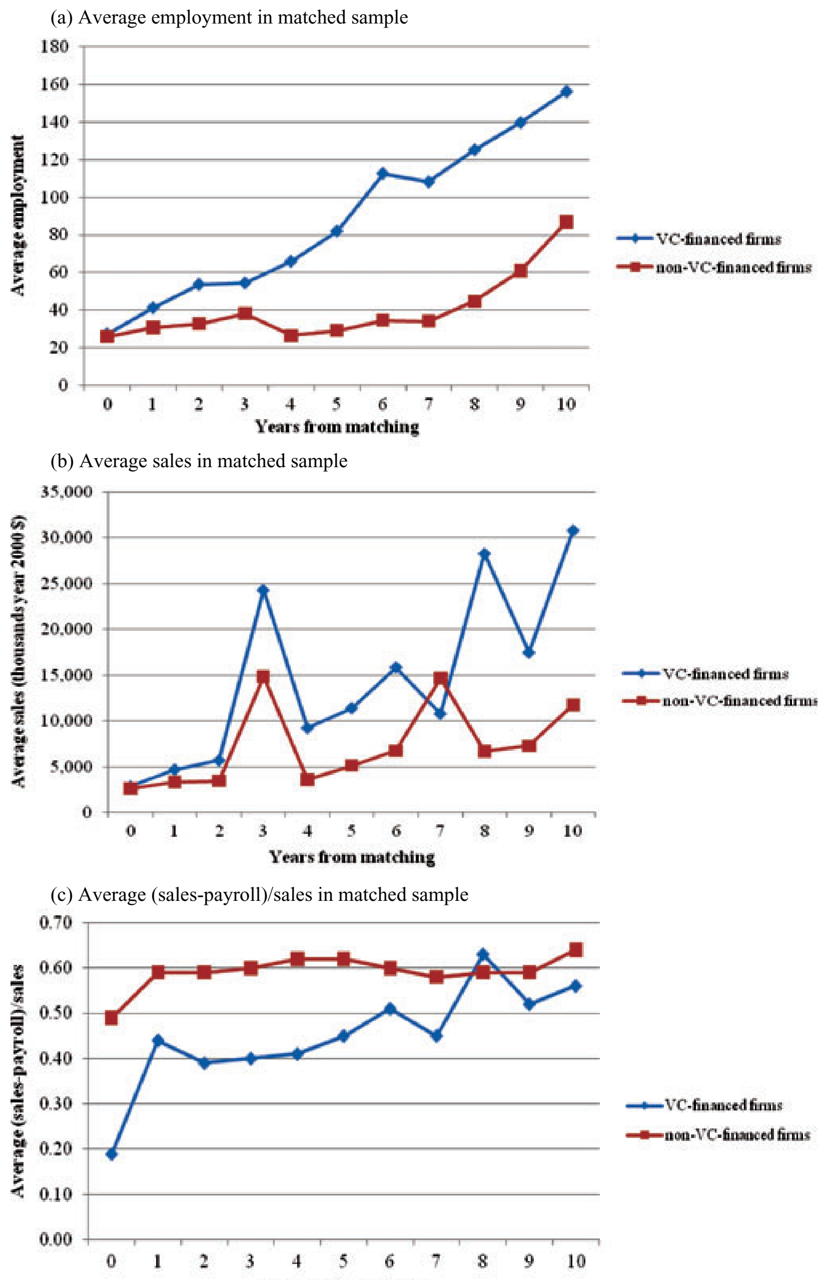

Figure 2. Average firm size and profitability in one-to-one matched sample. The sample is the one-to-one matched sample of 10,349 VC-financed and 10,349 non-VC-financed firms described in Appendix A. The firms are tracked from the time they enter the LBD to the year of their first exit event, if applicable, via failure, acquisition, or IPO. Employment and payroll data are taken from the LBD. Sales data are taken from the 1982, 1987, 1992, 1997, and 2002 waves of the Censuses of Services, Retail Trade, and Wholesale Trade and from the Longitudinal Research Database.

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firms also continue to grow on average, but at a much slower rate than the VC-financed firms even in this later point in their life cycles.[5]

Figure 2c plots payroll margin for VC- and non-VC-financed firms in match time. When VC-financed firms first receive VC, their payroll margins are dramatically lower than the matched non-VC-financed firms. The average payroll margin is 0.19 for VC-financed firms in the first year of VC finance while for matched non-VC-financed firms the average payroll margin is 0.49. Afterward, VC-financed firms' payroll margins hover around 0.40 and then gradually increase around 5 years after first VC financing. In contrast, non-VC-financed firms' payroll margins stay close to 0.60 in all years after matching. Figure 2c buttresses the claim that VCs invest heavily in employment, not only via larger numbers of employees but also via higher wages, in the first several years after investing in a firm. As firms age and exit, VC-financed firms' payroll margins come into line with those of non-VC-financed firms.

We also analyze the size differences between the matched VC- and non-VC-financed firms in a regression framework. We regress our firm size and profitability measures on a dummy variable, *VC*, that equals one for VC-financed firms as well as a time variable, *TimefromMatch*, that measures how far a firm is in years from being matched to its ex-ante observationally equivalent counterpart. We also interact *TimefromMatch* and the *VC* dummy to separately identify the time dimensions of each dependent variable for VC-financed firms. We run OLS panel regressions on *VC* and *TimefromMatch*, as well as the square of *TimefromMatch* to capture nonlinearities in the relation between firm size and time. In each regression we also include year fixed effects and industry fixed effects.

Table IV reports the coefficients and *t*-statistics, corrected for clustering by firm, for the OLS size and profitability regressions in our panel of matched VC- and non-VC-financed firms. The first three specifications regress the natural log of employment and sales as well as our payroll profitability measure on just the VC dummy variable. We see that VC-financed firms are on average larger, in terms of both employment and sales, and less profitable than non-VC-financed firms while VCs are involved with these firms as evidenced by the strongly significant coefficients on the VC dummy variable.

The last three specifications in Table IV allow us to see whether the growth pattern in employment, sales, and profitability differs for VC-financed firms over time. We find that VC-financed firms grow more quickly in terms of both employment and sales after VCs invest in them relative to their matched non-VC-financed counterparts, as evidenced by the positive and significant coefficients on *VC*TimefromMatch* in the first two regressions. However, the growth

[5] Because we observe sales infrequently in the Economic Censuses, taken every 5 years, and more frequently for the larger manufacturing firms in the LRD, we checked to see how firms that have non-missing sales data differ from those that do not have sales data in the matched sample. We find that firms with non-missing sales data are slightly larger and more successful than those with missing sales data. However, this is true for both VC- and non-VC-financed firms. Thus, the qualitative differences between VC- and non-VC-financed firm sales are not likely driven by sample selection bias.

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Table IV

One-to-One Matched Sample Firm Size and Profitability Regressions

The sample is the one-to-one matched sample of 10,349 VC-financed and 10,349 non-VC-financed firms. The sample tracks firms from their year of matching, that is, the year in which a VC-financed firm first receives VC, to the year prior to the first exit event, if applicable, via failure, acquisition, or IPO. Employment and payroll data are taken from the LBD. Sales data are taken from the 1982, 1987, 1992, 1997, and 2002 waves of the Censuses of Services, Retail Trade, and Wholesale Trade and from the Longitudinal Research Database. The large number of missing firm-years in the regressions for which *Log(Sales)* and *(Sales-Payroll)/Sales* are the dependent variables is due to the fact that sales data are observed in 5-year intervals except for the subset of firms in the LRD. *VC* is a dummy variable equal to one for VC-financed firms (= 1 in all firm-years for the VC-financed firm). *TimefromMatch* measures time in years from the point at which VC-financed and non-VC-financed firms are matched to each other. VC-financed firms are matched to non-VC-financed firms in the year they first receive VC financing. All regressions are estimated using OLS. Included in each regression are year fixed effects and industry fixed effects based on the industries defined in Table II. OLS coefficients are reported followed by t-statistics adjusted for clustering by firm in parentheses. *** and ** indicate statistical significance at the 1% and 5% level, respectively.

	Log(Employment)	Log(Sales)	(Sales-Payroll)/Sales	Log(Employment)	Log(Sales)	(Sales-Payroll)/Sales
VC	0.572***	0.895***	−0.179***	0.023	0.502***	−0.177***
	(30.34)	(28.09)	(−15.24)	(1.12)	(11.03)	(−6.67)
VC*TimefromMatch				0.250***	0.211***	−0.002
				(22.79)	(11.23)	(−0.16)
VC*TimefromMatch^2				−0.012***	−0.012***	0.000
				(−12.07)	(−8.73)	(0.45)
TimefromMatch				0.047***	0.121***	0.025***
				(7.73)	(10.11)	(5.57)
TimefromMatch^2				−0.001**	−0.003***	−0.001***
				(−2.03)	(−4.12)	(−4.26)
Industry fixed effects?	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects?	Yes	Yes	Yes	Yes	Yes	Yes
N	101,936	17,885	17,885	101,936	17,885	17,885
R^2	6.3%	12.8%	4.7%	12.3%	18.0%	5.0%

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rates in size for VC-financed firms also level off more rapidly in later years, perhaps as VCs exit their successful investments more rapidly, as evidenced by the negative and significant coefficients on $VC^{*}TimefromMatch^{2}$. Finally, the payroll margin regression suggests that VC-financed firms are initially less profitable, in terms of payroll margins, than non-VC-financed firms and continue to be so till exit.

The estimates in Table IV show that the patterns in employment, sales, and profitability between VC- and non-VC-financed firms that we observe in Figures 2a, 2b, and 2c hold in a regression framework. Thus, a key difference between VC-financed and non-VC-financed firms that emerges from our analysis is firm scale. Larger firm scale rather than higher profitability seems to be an important criterion for VC-financed firms to achieve prior to exit.

III. VC Financing and Firm Exit

We have seen that VC-financed firms change enormously relative to non-VC-financed firms in terms of size after venture capitalists become involved. However, it is unclear whether these differences tend to emerge because all VC-financed firms grow more quickly or because venture capitalists exit smaller firms more quickly relative to non-VC investors. One characterization of venture capitalists often found in anecdotal evidence is that they encourage the development of the one or two very high growth firms in their portfolio, that is, the potential eBays and Googles, and care little about the rest of their portfolio. Some argue that venture capitalists are quick to shut down companies; others suggest that venture capital is patient money and venture capitalists recognize the option value in their investments and exert effort to ensure companies do not close down. The nature of failure is arguably one of the least understood aspects of VC-financed firms. Because the LBD allows us to observe firm failures, we are able to present alternative statistics on VC-financed firm exits compared to the standard industrial data sources on VC-financed firms, such as VentureXpert and Venture Source. We are also able to compare these exit statistics to those of non-VC-financed firms.

A. Cumulative Exit Rates

Table V presents cumulative exit rates for all VC and non-VC-financed firms in the LBD. The exit rates are reported as of 2005, when our version of the LBD ends. The first set of columns in Table V presents exit percentages for firms that are created during the full sample period, 1981–2005. For this sample, we see that 34.1% of VC-financed firms fail, 26.4% get acquired, and 7.4% go public. For non-VC-financed firms, 66.3% fail, 0.85% get acquired, and 0.02% go public. Note that these percentages do not sum to 100% for each set of firms since some firms experience no exit event, that is, they continue as of 2005.

The second set of columns in Table V presents exit percentages for firms that are created between 1981 and 1997, 8 years before our sample period ends in 2005. Calculating exit rates for this set of firms results in a smaller

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Table V

Exit Rates of VC- and Non-VC-Financed Firms

The sample consists of all firms entering the LBD between 1981 and 2005. The first set of columns presents exit rates as of 2005 for all non-VC-financed firms that enter the LBD between 1981 and 2005 and for all VC-financed firms that enter the LBD and that first receive VC between 1981 and 2005. The second set of columns presents exit rates as of 2005 for all non-VC-financed firms that enter the LBD between 1981 and 1997 and for all VC-financed firms that enter the LBD and that first receive VC between 1981 and 1997. A firm is classified as having failed if it disappears from the LBD in its entirety, that is, all of its establishments are shut down. A firm is classified as acquired if it is classified in the LBD as having an ownership change in which it becomes part of another existing firm or if it can be matched to an M&A target in the SDC Platinum Mergers and Acquisitions database or in a Lexis/Nexis search. A firm is classified as having had an IPO from the SDC Platinum Global New Issues database. *** indicates statistical significance at the 1% level.

	Firms Created 1981–2005			Firms Created 1981–1997		
	VC-financed firms ($n = 17{,}763$)	Non-VC-financed firms ($n = 16{,}304{,}287$)	VC vs. non-VC z-statistic	VC-financed firms ($n = 7{,}037$)	Non-VC-financed firms ($n = 10{,}811{,}883$)	VC vs. non-VC z-statistic
% fail as of 2005	34.1	66.3	−86.1***	39.7	78.9	−88.2***
% acquired as of 2005	26.4	0.85	350***	33.5	1.04	290***
% IPO as of 2005	8.5	0.02	650***	16.1	0.02	750***

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percentage of firms that experience no exit event since we observe firms in the data for a period at least as long as the typical VC investment. These exit rates are informative since they measure longer-term exit rates of VC- and non-VC-financed firms. We see that 39.7% of VC-financed firms fail, 33.5% get acquired, and 16.1% go public. In contrast, 78.9% of non-VC-financed firms fail, 1.04% get acquired, and 0.02% go public.

We focus the rest of our analysis on the full 1981–2005 sample period and later account for the fact that firms may experience no exit event in multinomial logit models when comparing the exit patterns of VC- and non-VC-financed firms. We do this because it is important to include the latter part of our sample beginning in 1998, which experienced the biggest boom and bust cycle in the VC industry, when considering differences between VC- and non-VC-financed firms. Moreover, we later separately examine the boom period, 1998–2000, when examining VC- and non-VC-financed firm exit patterns.

We next examine in further detail the dynamics of firm exit. Table VI, Panel A, presents cumulative exit rates as of 2005 by firm age for all VC- and non-VC-financed firms created between 1981 and 2005 in the LBD. The cumulative failure rate of non-VC-financed firms 5 years after entry is 50.9%, while for VC-financed firms it is only 24.1%. Ten years after entry, the cumulative failure rate is 61.6% for non-VC-financed firms and 31.6% for VC-financed firms. These percentages at each age are based on the total number of firms in each sample. In the case of VC-financed firms, the cumulative exit percentages for each age are based on the full sample of 17,763 firms, and in the case of non-VC-financed firms the cumulative exit percentages at each age are based on the full sample of 16,304,287 firms.[6]

In Table VI, Panel A, we see that the large difference in cumulative failure rates between VC- and non-VC-financed firms is largely driven by much lower probabilities of VC-financed firms failing in the first years of life. At age two, the marginal probability of failing for VC-financed firms is 5.3% and for non-VC-financed firms is 13.9%, a difference of 8.6 percentage points. At age three, the marginal probability of failing is 6.1% for VC-financed firms and 8.7% for non-VC-financed firms, a difference of 2.6 percentage points. At age five, there is a one percentage point difference in marginal failure probabilities between VC- and non-VC-financed firms (3.4% versus 4.4%). After age five, the difference in marginal failure rates between VC- and non-VC-financed firms

[6] We keep the sample of firms fixed when calculating the cumulative exit percentages for each firm age in Table VI because this allows us to see in a non-parametric setting the dynamics of exit in the full sample of VC- and non-VC-financed firms that will form the basis of our parametric multinomial logit analysis in Section III.B. Since a large number of firms in our sample are created between 1996 and 2000, this means that the exit percentages reported at ages above 5 years are smaller than if we calculated the percentages only considering the firms that we can observe for at least the number of years specified in a given age. While it would be informative to also report cumulative exit percentages based only on the samples of VC- and non-VC-financed firms that we can observe at each age, we are unable to do so due to Census Bureau restrictions on the sizes of the samples underlying our reported statistics.

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Table VI
Full Sample Cumulative Exit Rates by Firm Age and Time from First VC Financing or Matching

Panel A reports cumulative exit rates as of 2005 by firm age for all non-VC-financed firms that enter the LBD between 1981 and 2005 and for all VC-financed firms that enter the LBD and that first receive VC between 1981 and 2005. Firm age is the number of years a firm is in the LBD. The percentages for each firm age are based on the full sample of VC-firms (n = 17,763) and non-VC-financed firms (n = 16,304,287) that enter the LBD between 1981 and 2005. Panel B reports cumulative exit rates for the one-to-one matched sample of 10,349 VC-financed and 10,349 non-VC-financed firms. In Panel B, cumulative exit rates as of 2005 are reported by years from the time that firms are matched to their non-VC-financed counterparts, that is, the first year of VC financing. The percentages for each year from matching are based on the full sample of VC- and non-VC-financed firms (n = 10,349) in the one-to-one matched sample. A firm is classified as having failed if it disappears from the LBD in its entirety, that is, all of its establishments are shut down. A firm is classified as having been acquired if it is classified in the LBD as having an ownership change in which it becomes part of another existing firm or if it can be matched to an M&A target in the SDC Platinum Mergers and Acquisitions database or in a Lexis/Nexis search. A firm is classified as having an IPO if it can be matched to a firm listed as having had an IPO in the SDC Platinum Global New Issues database. The exit event that occurs first, if applicable, is the exit event assigned to a firm. Two-tailed z-statistics testing for the equality of the cumulative exit rates between VC- and non-VC-financed firms are reported at the bottom of each panel. *** indicates statistical significance at the 1% level. — indicates that the number of firms underlying the calculation is too small to be disclosed.

Panel A. Firms in the LBD Created between 1981 and 2005

	Firm Age in Years									
	1	2	3	4	5	6	7	8	9	10
VC-financed firms (n = 17,763)										
% fail	4.4	9.7	15.8	20.7	24.1	26.7	28.5	29.9	30.9	31.6
% acquired	4.6	8.2	11.8	15.2	17.8	19.6	21.1	22.4	23.2	23.8
% IPO	0.8	1.7	3.1	4.4	5.3	6.1	6.6	7.0	7.4	7.6
non-VC-financed firms (n = 16,304,287)										
% fail	17.8	31.7	40.4	46.5	50.9	54.2	56.8	58.8	60.3	61.6
% acquired	0.2	0.4	0.5	0.5	0.6	0.6	0.7	0.7	0.7	0.8
% IPO	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01	0.01
z-statistic for % fail	−55.8***				−67.5***					−77.7***
z-statistic for % acquired	150***				280***					330***
z-statistic for % IPO	430***				520***					640***

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Panel B. One-to-One Matched Sample										
	Years from First VC Financing or Matching									
	1	2	3	4	5	6	7	8	9	10
VC-financed firms (n = 10,349)										
% fail	4.9	12.8	20.9	26.3	29.1	31.0	32.2	32.9	33.4	33.7
% acquired	5.6	10.4	15.0	18.0	20.6	22.0	23.0	23.7	24.0	24.2
% IPO	1.1	2.5	4.1	5.1	5.9	6.4	6.7	6.9	7.1	7.2
non-VC-financed firms (n = 10,349)										
% fail	9.5	21.8	30.8	36.8	40.1	42.4	43.9	45.0	45.6	46.3
% acquired	1.4	2.4	3.2	3.8	4.2	4.6	4.9	5.0	5.1	5.3
% IPO	—	—	—	—	—	—	—	—	—	0.14
z-statistic for % fail	−8.5***				−16.7***					−18.5***
z-statistic for % acquired	29.8***				35.8***					38.3***
z-statistic for % IPO	—				—					27.0***

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declines dramatically as firms continue to age.[7] This suggests that VCs make the biggest difference in the early years of firms' life cycles, or that they select firms that are much less likely to fail early on.

Turning to the cumulative acquisition and IPO rates in Table VI, Panel A, we see that VC-financed firms are much more likely to be acquired and go public relative to non-VC-financed firms. By age five, 17.8% of the VC-financed firms have been acquired and 5.3% have gone public, versus 0.6% and 0.01% for non-VC-financed firms. By age 10, these figures are 23.8% and 7.6% for VC-financed firms and 0.8% and 0.01% for non-VC-financed firms. This suggests that VCs actively promote or select their companies to exit via these exit routes earlier, perhaps by growing them more rapidly earlier in the life cycle.

In Table VI, Panel B, we report the cumulative exit rates of firms in our one-to-one matched sample as of 2005, rather than for the general population of VC- and non-VC-financed firms. We see that, even when we compare them to observationally equivalent non-VC-financed firms, VC-financed firms are once again less likely to fail and more likely to be acquired and to go public than non-VC-financed firms. Five years after receiving VC, 29.1% of VC-financed firms have failed, whereas 40.1% of matched non-VC-financed firms have failed. Further, 20.6% of VC-financed firms have been acquired and 5.9% have gone public after 5 years, whereas only 4.2% of matched non-VC-financed firms have been acquired and the percentage of non-VC-financed firms going public is too small to disclose. Ten years after first receiving VC, 33.7% of VC-financed firms and 46.3% of matched non-VC-financed firms have failed. Likewise, 24.2% of VC-financed firms have been acquired and 7.2% have gone public, whereas only 5.3% of matched non-VC-financed firms have been acquired after 10 years and only 0.14% have gone public.

B. *Timing of Exits*

Tables V and VI demonstrate that VC financing is strongly associated with a lower cumulative probability of failure. Thus, the larger VC-financed firm sizes relative to non-VC-financed firms that we observe in Section II.B are not driven by higher failure rates of VC-financed firms relative to non-VC-financed firms, so that only larger firms survive. Rather, VC-financed firms on average dominate non-VC-financed firms in terms of having both higher growth rates and lower failure rates even in our matched sample.

We next examine the timing of exits in more detail and ask whether the timing of firm failures, in particular, is different for VC- and non-VC-financed firms. Venture capitalists might push their companies hard to grow quickly, deciding relatively rapidly which firms have the best chance of achieving a

[7] We calculate marginal probabilities of failure at a given age by subtracting the cumulative failure probability for firms 1 year younger from the cumulative probability at the given age. For example, at age two, the marginal probability of failure for VC-financed firms is 9.7% minus 4.4%, or 5.3%, and the marginal probability of failure for non-VC-financed firms is 31.7% minus 17.8%, or 13.9%.

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successful exit and terminating those that do not in the interest of allocating more capital to the likely winners in their portfolios. Alternatively, VCs may have a more difficult time judging which firms will be successful in the early stages of investment and equally nurture and invest in all of their firms over a certain period of time.

We first focus on the failure rates reported for the matched sample in Table VI, Panel B. Examining the marginal failure rates of VC- and non-VC-financed firms by subtracting cumulative failure rates in successive years, we see that the largest differences in failure rates occur in the first few years after firms first receive VC. Two years after receiving VC, the marginal probability of failure is 7.9% for VC-financed firms and 12.3% for matched non-VC-financed firms, a 4.4 percentage point difference statistically significant at the 1% confidence level. Four years after receiving VC, the marginal probability of failure is 5.4% for VC-financed firms and 6.0% for matched non-VC-financed firms, a 0.6 percentage point difference, statistically significant only at the 10% confidence level. By 6 years after receiving VC, the marginal failure rate of VC-financed firms is 0.4 percentage points lower than that for matched non-VC-financed firms, 1.9% versus 2.3%, with statistical significance only at the 10% confidence level. The difference in marginal failure probabilities for VC- and non-VC-financed firms in the matched sample further decreases in later years and loses statistical significance.

We next examine whether these patterns in the timing of failures persist when we control for differences in industry and year. Since each firm in our one-to-one matched sample can experience only one exit event, we model firm exit in a multinomial panel logit model in which the excluded outcome is no exit, that is, the firm continues as of 2005. We report marginal probabilities calculated at sample means and z-statistics corrected for clustering by firm in parentheses for two multinomial logit specifications in Table VII. In the first specification we model firm exit as a function of a VC dummy and industry and year fixed effects. In the second specification, we also control for the number of years from being matched, *TimefromMatch*, as well as $TimefromMatch^2$ and interact these variables with the VC dummy to distinguish differences in exit probabilities between VC- and non-VC-financed firms over time.

The exit patterns we observe in Table VI are borne out in the multinomial logit models in Table VII. Focusing on the first model, we see that VC-financed firms are much more likely to be acquired and go public and are less likely to fail than non-VC-financed firms. On average, as per the marginal probabilities, VC-financed firms are 4.1 percentage points more likely to be acquired, 0.8 percentage points more likely to go public, and 1.3 percentage points less likely to fail in a given year.

In the second model, in which we allow for differences in the exit dynamics between VC- and non-VC-financed firms, we see that the probability that a VC-financed firm is acquired increases each year after the firm first receives VC-financing relative to non-VC-financed firms, as evidenced by a positive and statistically significant coefficient on the interaction between the VC dummy and *TimefromMatch*. The probability of going public for VC-financed firms

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Table VII

One-to-One Matched Sample Exit Multinomial Logits

The sample is the one-to-one matched sample of 10,349 VC-financed and 10,349 non-VC-financed firms. The table presents marginal probabilities, followed by z-statistics adjusted for clustering by firm, for two multinomial logit models. The base outcome in both models is no exit event for the firm, that is, the firm continues as of 2005. A firm is classified as having failed if it disappears from the LBD in its entirety, that is, all of its establishments are shut down. A firm is classified as having been acquired if it is classified in the LBD as having an ownership change in which it becomes part of another existing firm or if it can be matched to an M&A target in the SDC Platinum Mergers and Acqusitions database or in a Lexis/Nexis search. A firm is classified as having had an IPO if it can be matched to a firm listed as having had an IPO in the SDC Platinum Global New Issues database. The exit event that occurs first, if applicable, is the exit event assigned to a firm. *VC* is a dummy variable equal to one for VC-financed firms (= 1 in all firm-years for the VC-financed firm). *TimefromMatch* measures time in years from the point at which VC-financed and non-VC-financed firms are matched to each other based on ex ante observables. Note that VC-financed firms are matched to non-VC-financed firms in the year they first receive VC financing. Included in each multinomial logit are year fixed effects and industry fixed effects (based on the nine industries defined in Table II). The models are estimated using maximum likelihood. *** and ** indicate statistical significance at the 1% and 5% level, respectively.

	Multinomial Logit #1 (Base = No Exit)			Multinomial Logit #2 (Base = No Exit)		
	Fail	Acquired	IPO	Fail	Acquired	IPO
VC	−0.013***	0.041***	0.008***	−0.053***	0.028***	0.008***
	(−6.10)	(36.61)	(15.91)	(−15.35)	(13.49)	(6.17)
VC*TimefromMatch				0.018***	0.004***	−0.0001
				(10.04)	(6.06)	(−0.50)
VC*TimefromMatch^2				−0.001***	−0.0004***	0.0000
				(−6.09)	(−5.43)	(0.10)
TimefromMatch				0.009***	0.002***	0.0005***
				(10.07)	(4.07)	(2.96)
TimefromMatch^2				−0.001***	−0.0002***	−0.00003**
				(−10.26)	(−3.41)	(−2.56)
Industry fixed effects?	Yes	Yes	Yes	Yes	Yes	Yes
Year fixed effects?	Yes	Yes	Yes	Yes	Yes	Yes
N	105,031	105,031	105,031	105,031	105,031	105,031
Pseudo R^2	6.1%	6.1%	6.1%	7.6%	7.6%	7.6%
Wald Chi^2 (p-value)	4,561.9 (0.00)	4,561.9 (0.00)	4,561.9 (0.00)	5,561.1 (0.00)	5,561.1 (0.00)	5,561.1 (0.00)

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does not increase over time relative to that for non-VC-financed firms, as evidenced by a statistically insignificant coefficient on the interaction between the VC dummy and *TimefromMatch*. The probability of going public and getting acquired increases each year for both VC- and non-VC-financed firms as evidenced by a positive and statistically significant coefficient on *TimefromMatch*, but the increase slows over time as evidenced by the negative and statistically significant coefficient on $TimefromMatch^2$.

In the second model of Table VII, we also see that VC-financed firms have lower probabilities of failure overall, as evidenced by the negative and statistically significant coefficient on the VC dummy. All firms have an increasing marginal probability of failure over time, as evidenced by the positive and statistically significant coefficient on *TimefromMatch*. However, over time the failure probability for VC-financed firms increases more than that for non-VC financed firms as evidenced by the positive and significant coefficient on the interaction between the VC dummy and *TimefromMatch*.[8]

The estimates in Table VII suggest that VC helps keep firms alive in the early part of their life cycle. In the first several years after receiving VC, VC-financed firms typically grow rapidly in terms of employment and sales relative to non-VC-financed firms and have lower failure rates relative to matched non-VC-financed firms. However, after this initial growth period, the differences in marginal failure rates taper off. It is also possible that companies that survive for several years without VC financing are particularly strong companies, whereas it may take less for companies with VC financing to survive for this long. This evidence is consistent with VC allowing firms to grow and reach certain milestones before they are shut down.[9]

C. Do VCs Have Different Thresholds for Failure?

VC-financed firms are initially more likely to survive but then have similar failure rates relative to non-VC-financed firms. A question that arises is whether the threshold for firm survival is different for VC-financed firms. Do venture capitalists simply wait to see if there is option value to be realized, but then when they do shut firms down, have a more stringent criterion for what it means to be a successful surviving firm relative to investors in non-VC-financed firms?

To shed light on this question, we examine whether VC-financed firms look different from non-VC-financed firms at failure. We report average employment, sales, and payroll margins of firms in the year before they fail in Panel A

[8] When we compute the equivalent of the marginal probabilities of failure in the multinomial logit in the univariate statistics (as a function of firms that have not yet exited), we find that 5 years after VC financing, the marginal probability of failure for both VC- and non-VC-financed firms is 5.5%, and differences thereafter are not significant.

[9] VCs may allow firms to grow and reach milestones through different mechanisms. VCs may monitor the firm closely (Lerner (1995)), help alleviate financial constraints in the early part of firms' life cycles (for example, Clementi and Hopenhayn (2006), Cooley and Quadrini (2001), Desai, Gompers, and Lerner (2003)), and help professionalize the firm (Hellmann and Puri (2002)).

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of Table VIII. We see that VC-financed firms are on average larger than non-VC-financed firms when they fail, both in terms of employment and sales. However, they have lower payroll margins at failure. These results suggest that VCs grow all of their portfolio companies to a certain minimum level before deciding to shut them down, perhaps in an attempt to learn which of their portfolio companies will have successful acquisitions or IPOs.

D. Are VC-Financed Firm Failures Disguised as Acquisitions?

An alternative explanation for our results is that the lower failure rates for VC-financed firms reflect the fact that some failures are disguised as acquisitions. However, it may also be the case that both VC- and non-VC-financed firms have to meet minimum eligibility criteria before other corporations or public investors will buy their equity. To shed light on these alternative hypotheses, we compare the average size of VC- and non-VC-financed firms at acquisition and IPO in Panels B and C of Table VIII. There is no significant difference in the size of VC- and non-VC-financed firms at acquisition or IPO. Thus, it does not appear that venture capitalists are disguising failures as acquisitions, which implies that the lower overall failure rates for VC-financed firms truly reflect a difference in failure rates between VC-financed and non-VC-financed firms. Moreover, it appears that on average VC-financed firms are not able to take firms public or sell them to other firms without meeting size standards that non-VC-financed firms must also meet.

We also take an alternative approach to defining acquisition events in the LBD to test whether failures are being disguised as acquisitions. We define a "going concern" acquisition as an acquisition in which the acquired firm is not shut down within 2 years of the acquisition. Doing so re-classifies some acquisitions as failures. When we only count going concern acquisitions, 4,429 (or 43%) of VC-financed firms fail and 1,732 (or 17%) get acquired as going concerns, compared to 3,587 (or 35%) and 2,574 (or 25%) when we count all acquisitions, both going concern and non-going concern acquisitions. When we use the going concern definition of acquisitions, 5,175 (or 50%) of non-VC-financed firms fail and 421 (or 4%) get acquired, compared to 5,009 (or 48%) and 587 (or 6%) when we count both going concern and non-going concern acquisitions. Thus, using the going concern definition of acquisitions re-classifies many more VC-financed firms from acquisition to failure than it does non-VC-financed firms. However, the cumulative failure rates of VC-financed firms remain lower and their cumulative acquisition rates higher than for non-VC-financed firms.

We also examine the characteristics of VC- and non-VC-financed firms that fail and are acquired in the year before these two types of events using the going concern definition. Panel D of Table VIII presents average age and employment size for going concern acquisitions and failed firms. When compared to the statistics in Panels A and B of Table VIII, we see that the average size of firms at acquisition for "going concern" acquisitions is 61 and 64 employees for non-VC- and VC-financed firms, respectively, compared to 56 employees when we count all acquisitions. Thus, the firms acquired in going concern acquisitions

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Table VIII

Comparing VC- and Non-VC-Financed Firms at Exit

The sample is the one-to-one matched sample of 10,349 VC-financed and 10,349 non-VC-financed firms. Average values 1 year before the exit event are reported for VC- and non-VC-financed firms. Reported at the bottom of each panel are t-statistics for double-sided differences in the reported means. Employment and payroll data are taken from the LBD. Sales data are taken from the 1982, 1987, 1992, 1997, and 2002 waves of the Censuses of Services, Retail Trade, and Wholesale Trade and from the Longitudinal Research Database. Sales data are reported in thousands of 2000 dollars. The large number of observations with missing sales data is due to the fact that sales data are observed in 5-year intervals except for the subset of firms in the LRD. Panel A reports average employment, sales, and payroll margin for failed firms. A firm is classified as having failed if it disappears from the LBD in its entirety, that is, all of its establishments are shut down. Panel B reports average employment for acquired firms. A firm is classified as having been acquired if it is classified in the LBD as having an ownership change in which it becomes part of another existing firm or if it can be matched to an M&A target in the SDC Platinum Mergers and Acquisitions database or as identified by a Lexis/Nexis search. Panel C reports average employment for firms that have an IPO. A firm is classified as having had an IPO if it can be matched to a firm listed as having had an IPO in the SDC Platinum Global New Issues database. The exit event (i.e., failure, acquisition, or IPO) that occurs first, if applicable, is the exit event assigned to a firm. Panel D reports average employment for going concern acquisitions and for failed firms including non-going concern acquisitions re-classified as failures. A going concern acquisition is an acquisition in which all establishments are not shut within 2 years of the acquisition. ***, **, and * indicate statistical significance at the 1%, 5%, and 10% level, respectively. — indicates that the number of firms underlying the calculation is too small to be disclosed.

	Number	Age at Exit	Employment at Exit	Number with Sales Data	Sales at Exit	$\frac{\text{(Sales-Payroll)}}{\text{Sales}}$
Panel A. Failures						
VC-financed firms	3,587	4.8	26.9	818	3,103.0	0.5
non-VC-financed firms	5,009	4.6	15.1	935	1,250.6	0.7
t-statistic for VC vs. non-VC		3.14***	7.32***		7.53***	4.62***
Panel B. Acquisitions						
VC-financed firms	2,574	4.7	56.4	—	—	—
non-VC-financed firms	587	5.5	56.2	—	—	—
t-statistic for VC vs. non-VC		−4.27***	−0.02			

(*Continued*)

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Table VIII—*Continued*

Panel C. IPOs

	Number	Age at Exit	Employment at Exit	Number with Sales Data	Sales at Exit	(Sales-Payroll)/Sales
VC-financed firms	760	5.0	88.5	—	—	—
non-VC-financed firms	15	6.3	139.1	—	—	—
t-statistic for VC vs. non-VC		−1.72*	−0.61			

Panel D. Going Concern Acquisitions

	Going Concern Acquisitions			Failures (Including Non-Going Concern Acquisitions)		
	Number	Age at Exit	Employment at Exit	Number	Age at Exit	Employment at Exit
VC-financed firms	1,732	5.0	63.9	4,429	4.7	29.6
non-VC-financed firms	421	5.8	61.4	5,175	4.6	16.1
t-statistic for VC vs. non-VC		−3.60***	−0.29		1.42	9.29***

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are slightly larger than firms that are not. However, the difference in size between VC- and non-VC-financed firms is statistically insignificant even using the alternative going concern definition of acquisitions. Moreover, using the going concern definition of acquisitions, we find that VC-financed firms are larger at failure compared to non-VC-financed firms. Thus, the differences in size observed between VC- and non-VC-financed firms at exit are robust to considering going concern acquisitions.

E. Do Exit Patterns of VC-Financed Firms Differ by Time Period?

It is possible that the patterns we observe for the 1981–2005 sample period are driven primarily by the years prior to 2001. VC investing peaked in 1998–2000, which witnessed many IPOs in high-tech industries, but after which public equity market values declined dramatically. We ask whether the performance differential of VC-financed firms relative to non-VC-financed firms in terms of larger size, lower failure rates, and higher acquisition and IPO rates disappeared for firms that received VC in the so-called bubble period of 1998–2000.

To examine this question we select a sample of VC-financed firms that first received VC financing during 1998–2000 and their matched non-VC-financed firms. Many firms that received VC financing during this period may have been financed with the expectation of having a quick IPO, but the window for exit may have disappeared before these firms were ready to exit. In addition, it is possible that VCs backed lower quality firms during this period. We first examine the cumulative exit rates for VC- and non-VC-financed firms in this subsample. Table IX , Panel A, presents cumulative exit rates for the full sample period, 1981–2005, and for the subsample of firms receiving VC in the bubble period 1998–2000. For the subsample of firms receiving VC in 1998–2000, we see that the cumulative failure rate for VC-financed firms (42.7%) is less than that for non-VC-financed firms (51.5%); however, the difference is smaller than that for the full sample period (34.7% versus 48.4%). In addition, the acquisition rate for VC-financed firms is higher than that for non-VC-financed firms, as is the IPO rate. The IPO rate for VC-financed firms in the bubble period is much smaller than that for the entire sample period, reflecting in part the fact that IPO opportunities dried up shortly after firms in the bubble period first received VC financing.

Table IX, Panel B, shows the average sizes and ages of VC- and non-VC-financed firms that fail and are acquired in the year before they exit in the subsample of firms first receiving VC during 1998–2000. We see that the patterns we observe in the full sample in Table VII, Panels A and B, hold when we focus only on VC firms financed during the bubble period. In particular, failed VC-financed firms are larger and less profitable than failed non-VC-financed firms. Acquired VC-financed firms look similar to acquired non-VC-financed firms in terms of employment. One difference between the subsamples and the full sample, however, is that both VC- and non-VC-financed firms that fail are on average younger than firms that fail in the full sample.

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Table IX

Comparing VC- and Non-VC-Financed Firms' Outcomes for Firms Receiving VC during the "Bubble Period"

The sample is the one-to-one matched sample of 10,349 VC-financed and 10,349 non-VC-financed firms. Panel A presents cumulative exit rates as of 2005 in percentages. The first set of columns presents exit rates for the full sample period, that is, those VC-financed firms that receive their first round of VC between 1981 and 2005. The second set of columns presents exit rates for VC-financed firms that receive their first round of VC between 1998 and 2000 and for their matched non-VC-financed firms. A firm is classified as having failed if it disappears from the LBD in its entirety, that is, all of its establishments are shut down. A firm is classified as acquired if it is classified in the LBD as having an ownership change in which it becomes part of another existing firm or if it can be matched to an M&A target in the SDC Platinum Mergers and Acquisitions database or in a Lexis/Nexis search. A firm is classified as having had an IPO from the SDC Platinum Global New Issues database. Panel B reports average employment and sales at exit for firms that first receive VC or are matched to these VC-financed firms between 1998 and 2000. Sales data are taken from the 1982, 1987, 1992, 1997, and 2002 waves of the Censuses of Services, Retail Trade, and Wholesale Trade and from the Longitudinal Research Database. Sales data are reported in thousands of year 2000 dollars. The exit event (i.e., failure, acquisition, or IPO) that occurs first, if applicable, is the exit event assigned to a firm. *** indicates statistical significance at the 1% level. — indicates that the number of firms underlying the calculation is too small to be disclosed.

Panel A. Cumulative Exit Rates as of 2005 by Time Period

	Firms First Receiving VC 1981–2005			Firms First Receiving VC 1998–2000 ("Bubble Period")		
	VC-Financed Firms ($n = 10{,}349$)	Non-VC-Financed Firms ($n = 10{,}349$)	VC vs. Non-VC z-Statistic	VC-Financed Firms ($n = 3{,}944$)	Non-VC-Financed Firms ($n = 3{,}944$)	VC vs. Non-VC z-Statistic
% fail as of 2005	34.7	48.4	-20.1^{***}	42.7	51.5	-7.8^{***}
% acquired as of 2005	24.9	5.7	38.4^{***}	26.2	4.9	26.1^{***}
% IPO as of 2005	7.4	0.14	27.4^{***}	3.0	—	—

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Panel B. Comparing Firms First Receiving VC in 1998–2000 (Bubble Period) at Exit

Failures

	Number	Age at Exit	Employment at Exit	Number with Sales Data	Sales at Exit	(Sales-Payroll) Sales
VC-financed firms	1,683	3.8	27.7	362	3,144.1	0.4
non-VC-financed firms	2,029	3.4	16.7	324	1,136.0	0.6
t-statistic for VC vs. non-VC		6.71***	5.74***		5.27***	−3.14***

Acquisitions

	Number	Age at Exit	Employment at Exit	Number with Sales Data	Sales at Exit	(Sales-Payroll) Sales
VC-financed firms	1,034	3.9	48.1	—	—	—
non-VC-financed firms	194	3.6	38.2	—	—	—
t-statistic for VC vs. non-VC		1.48	1.57			

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Overall, the analysis indicates that the life cycle dynamics of VC- and non-VC-financed firms that we observe in the full sample of matched firms in the 1981–2005 period are qualitatively the same in the subsample of firms that first receive VC in the bubble period, 1998–2000. However, the outperformance by VC-financed firms in terms of lower failure rates and higher IPO and acquisition rates diminishes in this subsample of firms.

F. Are Exit Patterns of VC-Financed Firms Driven by Certain Kinds of VCs?

Another possible explanation for our results is that the relation between exit rates and VC financing, firm size, and profitability is driven by certain kinds of VC, for example, high or low reputed VC. A literature has developed that explores the impact of different types of VCs on investment outcomes (e.g., Botazzi, Hellmann, and da Rin (2008), Gompers (1996), Sorensen (2007), and Zarutskie (2010)). Understanding whether the relation between VC financing, firm scale and exits, both successes and failures, differs significantly for high and low reputed VCs is thus an interesting question in its own right.

We measure "high reputation" VC investors as those that are in the top quartile of the age distribution of VC firms in VentureSource and VentureXpert. There is evidence that older VCs and those that have more experience doing deals generally have higher returns (e.g., Sorensen (2007)). We use age as our measure of the reputation of a VC firm's quality. We find that our results are robust to alternative measures of VC reputation such as the number of past deals and number of past IPOs of the VC firm. We choose to report the results using VC age for the sake of brevity.

For each VC-financed firm in the LBD, we calculate the maximum age of the VC firms investing in the companies in their first round of VC financing. If the oldest VC firm that invests in the company's first round is in the upper quartile of the VC firm age distribution at the time it invests in the company, then the VC-financed firm is classified as having a high reputation VC firm as an investor. All other VC-financed firms are classified as receiving financing from low reputation VC investors.

Table X reports cumulative exit rates for VC-financed firms by VC investor reputation in the full sample and in the subsample of firms that first receive VC in the bubble period of 1998–2000. The cumulative exit rates for non-VC-financed firms are also presented. We see that, in both samples, VC-financed firms backed by high reputation VC investors have lower failure rates and higher acquisition and IPO rates than do VC-financed firms backed by low reputation VC investors. However, both high reputation and low reputation VC-financed firms have higher acquisition and IPO rates than do non-VC-financed firms. In addition, in the full sample and the subsample of firms first receiving VC between 1998 and 2000, both high reputation and low reputation VC-financed firms have lower failure rates than non-VC-financed firms. In fact, the difference in cumulative exit rates between low reputation VC-financed firms and non-VC-financed firms is statistically significant at the 1% level in both sample periods.

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Table X

Comparing VC- and Non-VC-Financed Firms Outcomes by VC Investor Reputation

The sample is the matched sample of 10,349 VC-financed and 10,349 non-VC-financed firms. The table presents cumulative exit rates as of 2005 in percentages. The first set of columns presents exit rates for the full sample period, 1981–2005. The second set of columns presents exit rates for VC-financed firms that receive their first round of VC between 1998 and 2000 and for their matched non-VC-financed firms. High reputation VC-financed firms are those whose oldest VCs in their first round of financing are in the top quartile of the VC age distribution. Low reputation VC-financed firms are those whose oldest VCs in their first round of financing are not in the top quartile of the VC age distribution. A firm is classified as having failed if it disappears from the LBD in its entirety, that is, all of its establishments are shut down. A firm is classified as acquired if it is classified in the LBD as having an ownership change in which it becomes part of another existing firm or if it can be matched to an M&A target in the SDC Platinum Mergers and Acquisitions database or in a Lexis/Nexis search. A firm is classified as having had an IPO in the SDC Platinum Global New Issues database. The exit event (i.e., failure, acquisition, or IPO) that occurs first, if applicable, is the exit event assigned to a firm. *** and ** indicate that the difference between High Reputation and Low Reputation VC percentages is statistically significant at the 1% and 5% level, respectively. ^^^ indicates that the difference between Low Reputation VC and Non-VC-Financed percentages is statistically significant at the 1% level. — indicates that the number of firms underlying the calculation is too small to be disclosed.

	Firms First Receiving VC 1981–2005			Firms First Receiving VC 1998–2000 (“Bubble Period”)		
	High Reputation VC-Financed Firms ($n = 1{,}614$)	Low Reputation VC-Financed Firms ($n = 6{,}049$)	Non-VC-Financed Firms ($n = 10{,}349$)	High Reputation VC-Financed Firms ($n = 523$)	Low Reputation VC-Financed Firms ($n = 2{,}190$)	Non-VC-Financed Firms ($n = 3{,}944$)
% fail as of 2005	29.6	36.7***	48.4^^^	36.9	44.7***	51.5^^^
% acquired as of 2005	27.8	24.8***	5.7^^^	35.2	24.5***	4.9^^^
% IPO as of 2005	8.9	7.3**	0.14^^^	4.8	2.3***	—

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IV. Conclusion

This paper is the first to our knowledge that uses a panel data set of the universe of employer firms in the United States over 25 years in conjunction with other government and proprietary data sources to empirically examine the life cycle dynamics of VC-financed and non-VC-financed firms. Using all firms across different industries and geographic areas as well as a matched sample of VC- and non-VC-financed firms, we explore differences in VC- and non-VC-financed firms to address several outstanding questions about the role of VC in new firm creation.

First, we examine the prevalence of VC in new firm creation in the economy. We see that, from the point of view of new firm foundings, VC-financed firms are an extremely small percentage of all new firms created—accounting for 0.11% over our 25-year sample period and increasing to 0.22% between 1996 and 2000. However, the picture is different if we focus on employment. The amount of employment generated by VC-backed firms accounts for 4.2% to 6.8% of employment in the United States between 1996 and 2000, steadily rising from 2.7% to 2.8% in the 1981–1985 period. Hence, in terms of employment generated, VC is more prevalent, and widely perceived to be an engine of growth for the economy.

We next ask what kinds of firms receive VC financing. We find that venture capitalists disproportionately invest in firms that have no commercial sales but that exhibit high levels of initial investment. Further, VC-financed firms grow larger than non-VC-financed firms, as measured by employment and sales, suggesting that VC backs firms that can achieve large scale. VC-financed firms also exhibit larger levels of sales but their payroll expenditures increase correspondingly, so VC-financed firms appear no more profitable than non-VC-financed firms before they are exited.

Since our data allow us to identify which firms fail, or are shut down, we generate statistics on exit percentages for VC-financed firms that cannot be obtained from the standard data sources on VC-financed firms. After the length of a typical VC investment, we find that 39.7% of VC-financed firms fail, 33.5% are acquired, and 16.1% go public. In contrast, 78.9% of non-VC-financed firms fail, 1.04% get acquired, and 0.02% go public.

We also analyze the exit dynamics of VC- and non-VC-financed firms, focusing in particular on failed firms. We find that the cumulative failure rate of VC-financed firms is lower than that of non-VC-financed firms. This difference is driven largely by lower failure rates in the initial years after receiving VC. We do not find evidence that the lower failure rate of VC-financed firms is driven by VC failures being disguised as acquisitions. We do find, however, that the performance difference between VC- and non-VC-financed firms diminishes in the post-internet bubble years, 1998–2000, although it does not disappear. We further examine particular kinds of VCs, for example, high reputation VCs, and do not find evidence that different types of VCs drive the results.

Our analysis of a large panel data set that contains both VC- and non-VC-financed firms allows us to better understand some of the fundamental features of VC-financed firms. Our findings also raise a number of future research

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questions. For example, while VC appears to have had a positive effect on firms over the past two decades, have the ways in which VC helps firm performance changed over time? If so, is this due to improved financial contracting or due to other interactive effects of VC?

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Appendix A. Identifying VC-Financed Firms in the Longitudinal Business Database and Creating a Matched Sample of VC- and Non-VC-Financed Firms

In this appendix we describe how we identify VC-financed firms in the Longitudinal Business Database (LBD) by matching it with information from VentureSource and VentureXpert, two proprietary databases on VC deals. We then describe how we create a one-to-one matched sample of VC- and non-VC-financed firms in the LBD using observable firm characteristics at the time the VC-financed firms receive their first round of VC financing. Finally, we investigate possible sample selection bias between matched and unmatched VC-financed firms in VentureSource and VentureXpert.

A. Identifying VC-Financed Firms in the LBD

We use the union of VentureSource and VentureXpert to identify the set of U.S. companies that receive VC financing between 1981 and 2005. Kaplan, Sensoy, and Stromberg (2002) discuss the relative merits of these two databases in recording VC deals. We use both databases to ensure the widest coverage of VC deals in the United States. Since comprehensive coverage of the VC industry by these databases does not begin until the 1980s, we focus our analysis on the period 1981–2005, the last year of the LBD, and focus our discussion on VC-financed firms that first receive VC between 1981 and 2005.

We identify 25,747 companies in the union of VentureXpert and VentureSource that first receive VC financing between 1981 and 2005. A little over half of the 25,747 companies are in both databases. Another 10% are in VentureSource only, and 40% are in VentureXpert only. We match 20,257 companies to the LBD for a raw match rate of 80%. According to Census Bureau researchers who regularly engage in name and address matching with Census micro-data sets, our 80% match rate is as high as what other researchers using these data sets have been able to attain, given the propensity for non-standardized spellings of words and occasional misspellings that appear in U.S. government administrative records data. We first attempt to match on a company's full name and full headquarters address, that is, city, state and zip code. This step results in 68% of the 20,257 matches we make. If a company matches to multiple firms in the LBD, we use the match that has the smallest difference between the first LBD year and the company founding year, or the year in which the company first receives VC financing if company founding year is missing. We next match companies using their full names and partial addresses, that is, city or zip code (but not both) and state, or state only. We identify 23% of our

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Table AI
Matching Criteria and Match Quality between VentureSource/VentureXpert and LBD

Panel A presents the distribution of firms in the union of VentureSource and VentureXpert that can be matched to the LBD by the matching criteria that matched them to the LBD. The first column presents the distribution for all matched VC-financed firms. The second column presents the distribution for VC-financed firms in the one-to-one matched sample of VC- and non-VC-financed firms. Panel B presents the average difference between VC-financed firms' first year in the LBD and their founding year in VentureSource and VentureXpert by matching criteria. The first column presents averages for all matched VC-financed firms. The second column presents averages for VC-financed firms in the one-to-one matched sample of VC- and non-VC-financed firms.

	All VC-Financed Firms in LBD ($n = 17{,}763$)	VC-Financed Firms in Matched Sample ($n = 10{,}349$)
Panel A. Distribution of VC-Financed Firms in LBD by Matching Criterion with VentureSource/VentureXpert		
Full Name, City, State, Zip	68.7%	70.9%
Full Name, either City or Zip, State	11.0%	10.7%
Full Name, State	12.0%	10.9%
Total	91.7%	92.5%
Full Name substring, City, State, Zip	5.0%	4.8%
Full Name substring, either City or Zip, State	1.6%	1.2%
Full Name substring, State	1.7%	1.5%
Total	8.3%	7.5%
Panel B. Average Difference between First LBD Year and VentureSource/VentureXpert Founding Year		
Full Name, City, State, Zip	0.8	0.6
Full Name, either City or Zip, State	0.8	0.7
Full Name, State	0.9	0.3
Full Name substring, City, State, Zip	0.9	0.2
Full Name substring, either City or Zip, State	0.6	0.7
Full Name substring, State	1.6	0.3

20,257 matches using full name and partial address. Combining these matches with the 68% made using full name and full address, we obtain 91% of our 20,257 matches using the full name of a company given in both VentureSource and VentureXpert and the LBD and some information from the company's headquarters address.

We make the final 9% of our matches using partial names of companies. Specifically, we match companies using substrings of their full names of length N, where N ranges from 20 to 10, and their full or partial address. This step allows us to match companies whose names might have different spellings in VentureSource and VentureXpert and LBD. We identify half of these matches using the company's full address and the other half using partial address information. As before, if a company matches to multiple firms in the LBD we

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use the match that has the smallest difference between the first LBD year and the company founding year, or the year in which the company first receives VC financing if the company founding year is missing.

Of the 20,257 matched firms, 17,763 firms first enter the LBD in 1981 or later and have non-missing industry information in the LBD. We use these 17,763 firms to form the basis of our analysis since we want to track firms from their first year of life in the LBD. The first column in Panel A of Table AI reports the percentage of the 17,763 VC-financed firms with non-missing industry data in the LBD that are matched according to the criteria described above. The majority (91.7%) of our VC-financed firms are matched using companies' full names and some information about their headquarters addresses. A sizable portion (68.7%) are matched using full company name and full company address.

To check the relative quality of the matches using partial company names and addresses as compared to matches based on full company names and addresses, we calculate the average difference between a company's first year in the LBD and its founding year as reported in VentureSource and VentureXpert and report this average difference by each matching criterion. The first column in Panel B of Table AI reports these averages. The average difference is similar across matching criteria, approximately 1 year (with a fairly narrow range of 0.8–1.6 years). This suggests that the quality of the matches using the less stringent criteria is not greatly diminished.

B. Creating a Matched Sample of VC- and Non-VC-Financed Firms

For each of the 17,763 VC-financed firms in the LBD, we identify non-VC-financed firms that share the same four-digit SIC code, age, geographical region, and number of employees as the VC-financed firms in the year that the VC-financed firms receive their first round of VC.[10] Some VC-financed firms have no non-VC-financed matches. Others have multiple non-VC-financed firm matches. We keep only VC-financed firms that have at least one non-VC-financed matched firm. If there are multiple non-VC-financed firms that match to the same VC-financed firm, we randomly select one of these firms as the VC-financed firm's matched counterpart. Likewise, if multiple VC-financed firms match to only one non-VC-financed firm, we randomly select a VC-financed firm to form the matched pair of VC- and non-VC-financed firms. This leaves us with a one-to-one matched sample of 10,349 unique VC-financed and 10,349 unique non-VC-financed firms.

We also form an alternative matched sample of VC- and non-VC-financed that does not require one-to-one matching. If at least one non-VC-financed

[10] Age is defined as the number of years in the LBD. The geographical regions are California and nine Census regions: New England (ME, VT, NH, MA, CT, RI), Mid-Atlantic (NY, PA, NJ), South Atlantic (DE, MD, WV, DC, VA, NC, SC, GA, FL), East North Central (WI, MI, IL, IN, OH), East South Central (KY, TN, MS, AL), West North Central (MN, IA, MO, ND, SD, NE, KS), West South Central (AR, OK, TX, LA), Mountain (MT, ID, WY, NV, UT, CO, AZ, NM), and Pacific (WA, OR).

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firm matches to a VC-financed firm, or vice versa, we keep all VC- and non-VC-financed firms with the same matching criteria, so that the number of VC-financed firms for a given set of match criteria may be greater or less than the number of non-VC-financed firms with the same match criteria. This leads to an unbalanced number of VC- and non-VC-financed firms in the alternative matched sample. We find that our results are robust to analysis of this matched sample. As described in Abadie and Imbens (2006), there are tradeoffs to estimating effects with these two different types of matched samples. We choose to be more conservative and report results using the smaller one-to-one matched sample.

The second column in Panel A of Table AI reports the percentage of the 10,349 VC-financed firms in the one-to-one matched sample that were matched to the LBD according to the name-and-address matching criteria described in Section A of this appendix. The percentage of firms that match according to each of the criteria is similar to that for the full sample of 17,763 VC-financed firms in the LBD. In particular, the majority of VC-financed firms in the one-to-one matched sample (70.9%) are matched using full name and full address. Another 21.6% match on full name and partial address, 4.8% are matched using partial name and full address, and 2.7% are matched on partial name and partial address.

The second column in Panel B of Table AI reports the average difference between the first year in the LBD and the VentureSource and VentureXpert founding year for VC-financed firms in the one-to-one matched sample. The average difference is between 0.2 and 0.7 years across the matching criteria, suggesting that there are no major differences in the quality of matches across criteria for the one-to-one matched sample of VC-financed firms. In addition, the match quality of the one-to-one matched sample appears slightly better than the overall match quality for the larger sample of 17,763 VC-financed firms.

Table AII presents characteristics of the 10,349 VC-financed firms in the one-to-one matched sample. The first panel shows that most firms (84%) receive VC within 3 years of entering the LBD. The second panel reports the number of employees that firms have in the year they receive VC. Most firms (80%) have 20 or fewer employees in the year they receive VC. The third panel reports the geographical regions in which firms are located. The largest percentage of firms (41%) is located in California. Another 36% are located on the east cost, in New England, the Mid Atlantic, or the South Atlantic. The remaining 23% are located in the interior of the United States or in the Pacific Northwest.

C. Investigating Possible Sample Selection Bias

Finally, we investigate the characteristics of the VC-financed firms we study relative to the VC-financed firms in VentureSource and VentureXpert that we are not able to match to the LBD. Note that there are two relevant samples here. The first comprises the 17,763 VC-financed firms in VentureSource and VentureXpert that we were able to match to LBD relative to 7,984 VC-financed firms in VentureSource and VentureXpert that we were not able to match to

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the LBD. The second sample comprises the 10,349 VC-financed firms in the one-to-one matched sample relative to the characteristics of the remaining 15,398 VC-financed firms in VentureSource and VentureXpert. Most of our results are based on the one-to-one matched sample, so for our purpose this is the most relevant comparison. However, for completeness, we report statistics for both samples.

Comparing the sample of 17,763 VC-financed firms matched to LBD relative to the remaining 7,984 VC-financed firms in VentureSource and VentureXpert, we find that in general the former firms tend to be more likely to go public (8.5%, as reported in Table V, versus 6%) and more likely to be acquired (26.4%, as reported in Table V, versus 18%). Hence, they are seemingly more successful firms; however, the IPO and acquisition rates of 6% and 18% for the unmatched firms are still much higher than those rates for non-VC-financed firms in the LBD, which are 0.02% in the full sample and 0.14% in the matched sample for IPOs and 0.85% in the full sample and 5.7% in the matched sample for acquisitions. Further, there is little difference in the failure rates as reported by VentureXpert and VentureSource for these two sets of firms. In fact, the reported failure rates are higher for the matched firms (20%) than for the unmatched firms (17%) according to VentureSource and VentureXpert. Note that these failure rates are those reported in VentureXpert and VentureSource and are therefore lower than those reported in Table V for VC-financed firms matched to the LBD. In addition, both sets of firms receive similar amounts of VC financing in the first round ($6 million) and similar cumulative total VC raised, $22 million for matched firms versus $18 million for unmatched firms and firms with missing industry data.

When we consider the one-to-one matched sample of 10,349 VC-financed firms relative to the characteristics of the remaining 15,398 VC-financed firms in VentureSource and VentureXpert, we find that the difference between these samples is even smaller. The one-to-one matched sample firms have slightly lower IPO rates, 7.4%, as reported in Table IX, Panel A, versus 8% for all other VC-financed firms, and slightly smaller first rounds of financing, $5 million versus $7 million, with the cumulative VC amount raised being similar at $21 million. VC-financed firms in the one-to-one matched sample also have higher acquisition rates, 24.9%, as reported in Table IX, Panel A, versus 22% for all other VC-financed firms. The reported failure rates in VentureXpert and VentureSource for the VC-financed firms in the one-to-one matched sample is also higher at 22% versus 16% for all other VC-financed firms. Overall, the VC-financed firms in the one-to-one matched sample do not appear any larger or more successful when compared to the VC-financed firms in VentureSource and VentureXpert that are not in the one-to-one matched sample. In fact, by some measures the one-to-one matched VC-financed firms appear to be smaller and less successful than these other VC-financed firms. This suggests that our performance results and the lower failure rates that we observe when we compare VC- and non-VC-financed firms in the matched sample are not biased by examining only the more successful firms in VentureSource and VentureXpert.

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Table AII

Characteristics of the One-to-One Matched Sample of VC- and Non-VC-Financed Firms

The sample includes VC-financed firms that enter the LBD between 1981 and 2005 that first receive VC financing in the year they enter the LBD or in any subsequent year and that can be matched to a non-VC-financed firm that is of the same age, in the same four-digit SIC code, in the same geographical region, and in the same employment size category in the year that the VC-financed firm has its first round of VC financing. The number and percentage of VC-financed firms in the sample that are matched at a particular age, employment size category, and geographical region are reported. Firm age measures the number of years a firm is in the LBD. A headquarters geographical region is defined as follows. New England is defined by the states ME, VT, NH, MA, CT, and RI. Mid Atlantic is defined by the states NY, PA, and NJ. South Atlantic is defined by the states DE, MD, WV, DC, VA, NC, SC, GA, and FL. East North Central is defined by the states WI, MI, IL, IN, and OH. East South Central is defined by the states KY, TN, MS, and AL. West North Central is defined by the states MN, IA, MO, ND, SD, NE, and KS. West South Central is defined by the states AR, OK, TX, and LA. Mountain is defined by the states MT, ID, WY, NV, UT, CO, AZ, and NM. Pacific Northwest is defined by the states OR and WA.

	Number of VC-Financed Firms	Percentage
Firm age at Matching		
1	5,246	51%
2	2,372	23%
3	1,059	10%
4	581	6%
5	331	3%
6	199	2%
7	160	2%
>7	401	4%
Total VC-financed firms	10,349	100%
Number of Employees at Matching		
1–3	2,386	23%
4–6	1,760	17%
7–10	1,760	17%
11–20	2,428	23%
21–30	996	10%
31–50	632	6%
51–70	168	2%
71–100	101	1%
>100	118	1%
Total VC-financed firms	10,349	100%
Region		
California	4,208	41%
New England	1,258	12%
Mid Atlantic	1,239	12%
South Atlantic	1,266	12%
East North Central	512	5%
East South Central	123	1%
West North Central	297	3%
West South Central	619	6%
Mountain	467	5%
Pacific Northwest	360	3%
Total VC-financed firms	10,349	100%

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Appendix B. Identifying Acquisitions in the LBD

In this appendix we describe how we identify acquisitions in the LBD. We first document establishment ownership changes using information contained in the LBD. We classify a firm as acquired if all of its establishments become part of another existing firm as recorded by the LBD. Ownership changes in the LBD are primarily tracked in the years in which the Economic Census is taken, when detailed questions about establishment ownership are asked. Since the Economic Census only occurs every 5 years, many ownership changes may not be recorded if acquired establishments are shut down before the next Economic Census is taken. We therefore use two additional external data sources to identify company acquisitions.

We merge the SDC Platinum Mergers and Acquisitions (SDC M&A) database to the LBD to identify acquisition targets in the LBD. We do this using name and address matching similar to that described in Appendix A. Because the SDC M&A database has greater coverage of larger acquisitions and acquisitions by public acquirers, we further augment our search for acquisitions using news articles published between 1981 and 2005 as obtained from Lexis/Nexis. We search all English language newspaper, magazine, newsletter, and news wire articles published between 1981 and 2005 related to M&A events in the United States for the names of all VC-financed companies and non-VC-financed companies in our one-to-one matched sample. Because we cannot remove the names of the VC- and non-VC-financed firms in our sample from the secure Census computer lab to conduct searches using the standard Lexis/Nexis interface, we import all such news articles published between 1981 and 2005 as text files and then search for articles related to companies in our Census sample using Perl code.

After identifying the articles related to the VC- and non-VC-financed firms in our Census sample, we then read each article and tag the articles that inform the reader whether the company in question had in fact been acquired. For companies that are reported to have been acquired, we record the date of the acquisition as reported by the news article. This search and data collection effort took a total of 6 months with the help of 25 research assistants.

We combine the information on acquisition events from our Lexis/Nexis searches with the acquisitions identified using the SDC M&A database and LBD-identified ownership changes to arrive at our final set of acquisitions. Without the Lexis/Nexis searches, the majority of acquisitions (around 70%) are identified using the SDC M&A database, highlighting how few acquisitions are recorded in the LBD itself. The acquisition rates are 20% for VC-backed firms and 4.3% for non-VC backed firms (as of 2005) when we just use the SDC M&A database and LBD-identified ownership changes. Adding the Lexis/Nexis searches increases the acquisition rates to 25% for VC-financed firms and 5.7% for non-VC-financed firms. It is important to note that these acquisition rates reflect right censoring in the data since many of the firms in our sample are started in the late 1990s and early 2000s and not recorded as acquired as of 2005 when our sample ends, even if they are acquired after 2005. If we

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restrict our sample to firms that first received VC financing before 1998 (or were matched to such a VC-financed firm), we find that 33% of VC-financed firms and 8.8% of non-VC-financed firms are acquired. These rates become even higher, between 35% and 40% for VC-financed firms, if we restrict our attention to firms that receive VC in 1995 or earlier. These figures are in line with observed acquisition rates of VC-financed firms of 30% to 50% 10 years after first receiving VC (for example, Sand Hill Econometrics database).

Using Lexis/Nexis news searches to identify acquisitions in the LBD in addition to the SDC M&A database does not have a meaningful impact on the difference in acquisition rates between VC- and non-VC-financed firms, though it does change the overall acquisition rates. Adding the searches increases the set of identified acquisitions for VC-financed firms by 24%, from 2,074 to 2,574 acquisitions. This corresponds to an increase in the total acquisition rate from 20.0% to 24.9%, out of 10,349 VC-financed firms. The Lexis/Nexis searches increase the set of identified acquisitions for non-VC-financed firms by 32%, from 446 to 587 acquisitions. This corresponds to an increase in the total acquisition rate from 4.3% to 5.7%, out of 10,349 non-VC-financed firms.

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